

Health Insurance and Health-Seeking Behavior: The Distributional Effects of Public Health Insurance in Nepal^{*}

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NOVEMBER 2025

Expanding access to health insurance is a major policy priority in low- and middle-income countries, yet rigorous evidence on its effects remains limited. This paper examines the impact of Nepal's National Health Insurance Program (NHIP), a voluntary public health insurance program. I exploit its staggered rollout as a natural experiment and use a difference-in-differences strategy to estimate causal effects of NHIP eligibility using novel administrative data. My estimates show that health insurance eligibility significantly increases health service use: total visits rise by 14.2 percent, outpatient visits by 13.7 percent, and emergency visits by 19.6 percent. Insurance eligibility more than doubles the use of laboratory services (160.7 percent) and increases radiographic services by 18.6 percent. Dynamic effects suggest gradual uptake, with significant impacts emerging only after five quarters of implementation. However, the benefits are not evenly distributed. Utilization gains are larger among older populations and in richer districts, while poorer districts experience little measurable effect. Gender effects are modest: both male and female visits increase substantially, with only a small relative shift toward male patients. Taken together, these findings highlight both the promise and the limitations of public health insurance in a lower-middle-income country context.

Keywords: Health Insurance, NHIP

JEL-Classification: H51, I12, I13, I14, J16, O15

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1 – Introduction

The development of a well-functioning health care system marks an essential step down the path of national economic development. A common roadblock facing lower-middle income countries in this progression is the presence of high out-of-pocket medical costs, which leave households financially vulnerable to health shocks (Chetty and Looney 2006). Accordingly, the expansion of health insurance is often a paramount priority for developing nations (Kurowski et al. 2023; Miller, Pinto, and Vera-Hernández 2013), with the establishment of Universal Health Coverage (UHC) as the ultimate objective. While previous authors have outlined the importance of UHC in improving both public health (Kurowski et al. 2023; WHO and Bank 2023) and health care utilization (Das and Do 2023; Fitzpatrick and Thornton 2019; Huang and Liu 2023; Miller, Pinto, and Vera-Hernández 2013), there remains no consensus as to the impact of health insurance on healthcare utilization in lower-middle income countries.

To address this absence, I study the launch of the National Health Insurance Program (NHIP), a Nepalese public health insurance program that aimed to prevent health-related expenditures driving households into poverty and achieve UHC in Nepal. It was initially piloted in three districts in 2016 and gradually expanded to all 77 districts by 2022. I study this rollout by compiling monthly data from all the districts from 2014 to 2022 on health care utilization. Overall, this unique dataset - the first of its kind compiled in Nepal - allows me to study the impact of health insurance on health care seeking behaviours. Further, I study the disparities in accessing health care across gender, age, and poverty dimensions.

I use differences in differences methodology to get the causal impact of eligibility on health insurance by exploiting this natural experiment where the insurance was rolled out in phases over the years, and employ the Callaway and Sant’Anna (2021) estimator to address methodological challenges in analyzing staggered policy rollouts, providing more credible causal estimates by accounting for treatment effect heterogeneity.

2 – Background

2.1. Country Context

Nepal is categorized under lower-middle income country by World Bank with a population of 29.2 million people (National Statistics Office 2023). It is a federal democratic republic with three tiers of government; federal, provincial and local. It is divided into 7 provinces and 77 districts. Gross National Income (GNI) per capita of Nepal reached USD 1380 in 2022 where 17% of total population are multidimensionally poor (World Bank 2025a; GoN 2021).

The health care system in Nepal is a mix of public and private providers, with the public sector providing the majority of health services. The public health care system is organized into three tiers: primary health care centers, district hospitals, and tertiary care hospitals. The private sector includes private hospitals, clinics, and pharmacies. The government provides free basic health services at public health facilities, but many people still rely on private providers for more specialized care. The OOP expenditure per capita of Nepal was USD 213.4 in 2022 which is more than 50% of total health expenditure (WHO 2022). The life expectancy at birth of Nepal is 70 years, which is slightly higher than the average of lower-middle income countries with 69 years (World Bank 2025b).

2.2. The National Health Insurance Program of Nepal

National Health Insurance Program (NHIP) is a public health insurance program implemented by the government of Nepal starting in 2016 with the major objective of reducing out-of-pocket (OOP) health expenditure of households. In 2017, Health Insurance Board (HIB) was established to implement health insurance nationwide with a mandate of covering every citizen under health insurance. Before NHIP there were few insurance programs from a couple of hospitals and some private providers but neither of them had any substantial effects on take up and use (Ayer et al. 2024). There were few programs run by Ministry of Health and Population (MoHP) which provided free treatment for essential non-communicable diseases like cardiovascular diseases and malignancy alongside Safe Motherhood Program (SMP) that provided free maternal/neonatal care to pregnant women. I discuss SMP in further detail in Section 6.

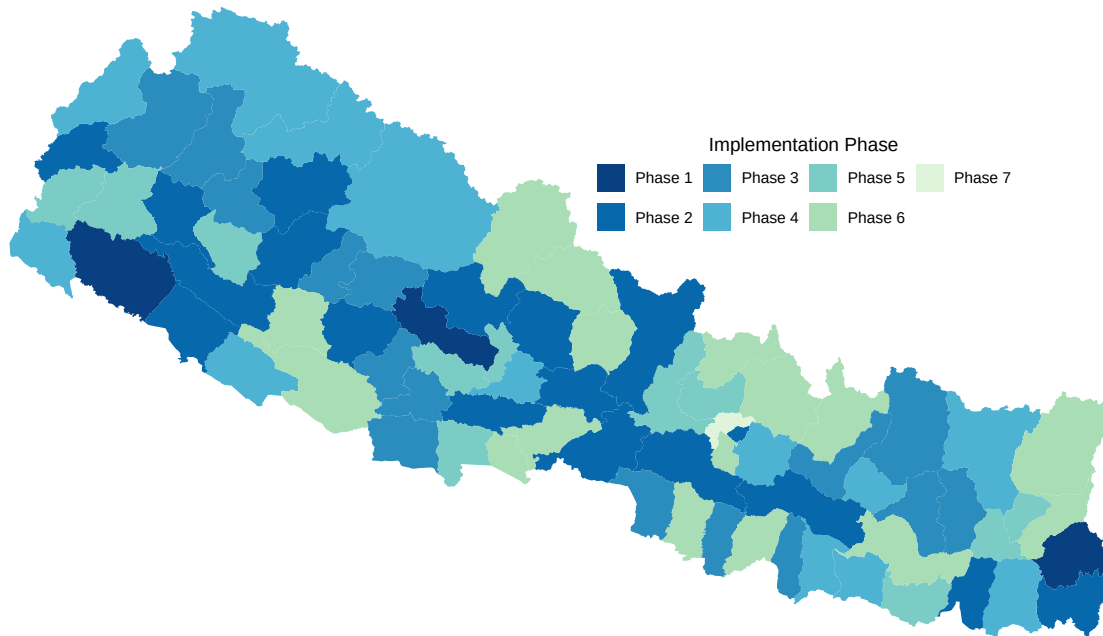
Figure 1 shows the phase wise implementation of health insurance in various districts from 2016

to 2022. NHIP is a voluntary insurance program, where it receives contribution from households and has to be renewed annually. It is primarily funded through government budget allocation and premiums collected from the enrolled. A family of five members can register for health insurance at the annual premium of NPR 3,500 (USD 25.23) ¹, with any additional member costing NPR 700 (USD 5.05). Once registered the members of NHIP are entitled to free care up to maximum of NPR 100,000 (USD 720.96) per family per year. Any additional members get coverage of NPR 20,000 (USD 144.19). Alongside this, government also provides subsidies to ultra-poor, senior citizens, disabled and patient of selected terminal diseases. Elderly members above the age of 70 with selected terminal diseases receive additional coverage of NPR 100000 (USD 720.96) (HIB 2024).

NHIP generously covers all preventive, curative, promotive, inpatient, emergency, surgery, medicines, diagnostics, rehabilitation and health aid equipments, alongside covering up to NPR 2000 (USD 14.42) of emergency ambulance service. It excludes services like expensive medical aid equipments, plastic surgery, most dental services and artificial insemination. Households during the registration select their preferred first point of contact (health care centers), which provides them facilities and if needed refers them to secondary care. It is a completely cashless system which does not require any payment at any point.

1. USD 1 = NPR 138.7 at the time of writing

Figure 1 – Coverage of Health Insurance in Nepal



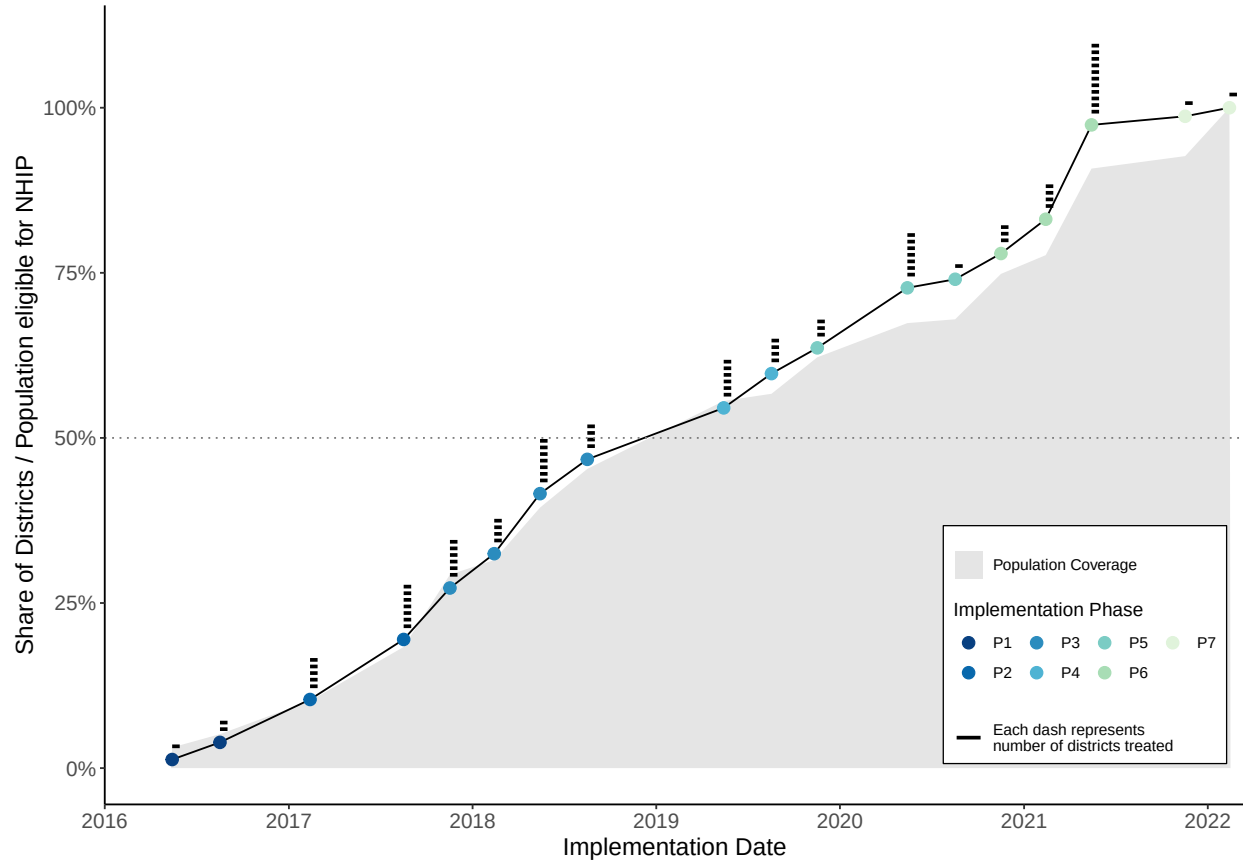
Source: Authors own calculations based on the data collected from annual reports, press releases and notices.

Notes. This figure illustrates the spatial variation of the coverage of National Health Insurance Program across different districts from 2016 to 2022

2.3. Rollout of NHIP

The rollout of the National Health Insurance Program (NHIP) was implemented in multiple phases at the district level which are second level of administrative division as shown in [Figure 1](#). [Figure 2](#) shows the rollout of the NHIP program from 2016 to 2022. The order of rollout of the insurance was not announced prior to implementation and implementation dates within the year were different due to administrative factors. Health Insurance Act was enacted in 2017 and Health Insurance Board (HIB) was formed after second phase of implementation. The first phase of implementation started in three districts in 2016, and gradually expanded to all 77 districts by the end of 2022.

Figure 2 – Rollout of the National Health Insurance Program



Source: Authors own calculations based on the data collected from annual reports, press released and notices.

Notes. This figure illustrates the implementation of the National Health Insurance Program across different districts from 2016 to 2022. Dash at each point indicates the number of districts that were newly covered by health insurance at that time.

In [Table 1](#), I present pre-treatment district characteristics across treatment timing cohorts. None of the differences are statistically significant, and standardized differences are uniformly small, with all values below 0.20 and most below 0.10, indicating a high degree of balance in observable characteristics and baseline health service utilization. This suggests that the timing of the health insurance rollout was not strongly correlated with observable district characteristics.

Table 1 – Pre-Treatment Characteristics by Treatment Timing Cohort

Variable	Mean (Early)	Mean (Late)	Diff.	Std. Diff.	P-value
<i>Demographics</i>					
Population (1000s)	388.43	408.51	20.09	0.06	0.80
Poverty incidence	0.29	0.27	-0.02	-0.16	0.49
No. of hospitals	111.11	114.92	3.81	0.08	0.72
Female population (1000s)	179	185.97	6.97	0.05	0.83
No. of households (1000s)	69.73	75.31	5.57	0.09	0.70
Female headed household share	0.07	0.07	0	0.02	0.92
Migrants share	0.27	0.25	-0.02	-0.15	0.53
Agricultural occupation share	0.84	0.83	-0.01	-0.09	0.70
<i>Dependent Variables</i>					
Total clients served (1000s)	70.57	81.75	11.17	0.15	0.51
TCS ages 0-9 (1000s)	14.28	14.73	0.45	0.04	0.86
TCS ages 10-19 (1000s)	14.71	15.34	0.62	0.06	0.80
TCS ages 20-59 (1000s)	32.97	41.34	8.37	0.19	0.39
TCS ages 60+ (1000s)	8.61	10.34	1.74	0.15	0.50
Female share of visits	0.58	0.58	-0.01	-0.19	0.40
No of Districts	37	40			

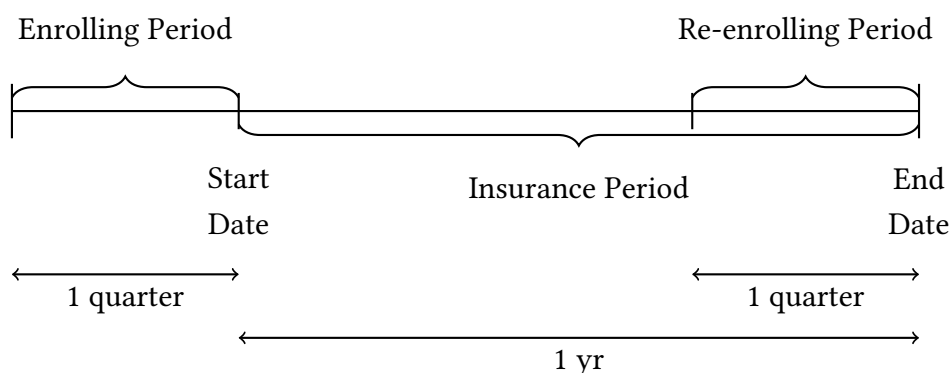
Source: Demographic variables are the authors' own calculations based on the National Census 2011.

Notes: The table reports district-level pre-treatment averages by treatment timing cohort. Differences are defined as later-treated minus earlier-treated districts. Standardized differences are calculated as mean differences divided by the pooled standard deviation. P-values are from Welch two-sample t-tests.

2.4. NHIP Procedures

The process of getting into NHIP follows a specific process as shown in Figure 3. There are four start dates every year which are February, May, August and November. Before the start date, 1 quarter is designated as enrolling period, where everyone who enrolls during that period have the same start date. And the renewal is only allowed 1 quarter before the end date of the insurance period. This process does not allow renewals during their insurance period, so everyone registered are limited to their annual insurance allowance.

Figure 3 – NHIP Enrollment Timeline



The enrollment assistants at each ward level collect the information about the households willing to register for health insurance, which is later verified at the district level which officially enrolled them into the program. Once registered, the members choose generally the closest public health care center as their primary service point of contact and receive their insurance card. The first point of contact provides all the primary health care services, and if needed refers them to secondary or tertiary care which can be both private and public service providers. The empaneled service providers are both public and private hospitals. At the end of 2022, about 33% of total households had at least one member enrolled in the NHIP program, though individual-level coverage rates are considerably lower as not all household members register.

3 — Data

3.1. *NHIP Rollout*

I track the implementation of health insurance through survey of annual health reports of Department of Health Services (DoHS), notices and press releases from HIB. I create a rollout data for each district including exact date of commencement of registration and the date of implementation of health insurance.

3.2. *District Health Information System*

I use comprehensive data from the District Health Information System (DHIS2) for this study. DHIS2 collects monthly data on visits, use of health services, referrals, family planning services, immunization and data related to diseases like malaria, leprosy, HIV/AIDS, tuberculosis among others. It collects data for both public and private health services. Reporting to DHIS2 was mandated for all public health facilities beginning in 2014, two years before NHIP's launch, ensuring that the reporting infrastructure was established before the program began. While reporting is not compulsory for the private sector, preceding the implementation of health insurance, robust data collection via DHIS2 was one of the government's priorities, and almost all empanelled healthcare institutions provide data through DHIS2.

Crucially, the district-level aggregated data used in this study are compiled from a consistent set of facilities that reported throughout the 2016–2022 study period, effectively constituting a balanced facility panel. This construction ensures that the estimated utilization changes reflect changes in patient visits at the same facilities over time, rather than the entry or exit of reporting units.

3.3. *Health Insurance Board Claims Data*

As a secondary data source, I use administrative claims records from the Health Insurance Board (HIB). The HIB records each insurance-financed visit at empanelled facilities, including the date, facility, patient demographics (age and gender), and district of residence. Unlike the DHIS2 data, which capture all visits regardless of insurance status, the claims data record only visits where the patient used their insurance benefits. I aggregate these records to the district–quarter level to match

the structure of the main analysis data. The claims data are available for treated districts from the quarter of NHIP implementation onward, covering the period up to 2022. I use these data in the validation exercise (Table 9) to compare the magnitude of DiD-implied utilization changes with the observed volume of insurance-financed visits, providing an independent check on the plausibility of the main estimates. Because claims data capture only insured visits, they are expected to be smaller than the DiD-implied total visit increases, which include both insurance-financed and non-insurance visits induced by the program.

3.4. Household Risk and Vulnerability Survey and Nepal Living Standards Survey

I use two of the living standards survey in Nepal, the Nepal Living Standards Survey (NLSS) 2010/11 and three year panel survey, Household Risk and Vulnerability Survey (HRVS) of 2016 to 2018 (CBS 2011; World Bank 2020). Both of the surveys follow Living Standards Measurement Survey (LSMS) methodology developed by World Bank. And they provide comprehensive household information on health expenditures, consumption expenditures, education expenditures and labor supply.

4 — Empirical Method

4.1. Empirical Strategy

My primary goal is to identify the causal effect of health insurance on various important outcomes. Simply comparing districts that implement health insurance to those who do not implement health insurance results in bias because those districts that implement health insurance might differ in many unobservable ways that also correlate with each of their potential outcomes.

To get away from this bias, I use the gradual rollout of the health insurance and use a Difference in Differences (DID) method for casual identification of the effect of health insurance program. This method exploits the variation in timing of implementation, with some districts implementing earlier than others. It compares the evolution of outcomes for the earlier treated groups with groups not yet treated, controlling for district time invariant confounders and time-specific effects that impact all districts.

4.2. Estimator

Prior literature has shown that the popular two-way fixed effects (TWFE) estimator, where the outcome is regressed with the group and time fixed effects, can be biased in a staggered timing setup because it imposes an assumption of homogenous effects in treatment timing and homogenous effects in groups. This homogeneity is unlikely because of various reasons including that districts that implement health insurance earlier might get efficient with time with better enrollment strategies and service delivery. If there is any treatment timing heterogeneity, the static TWFE estimator can be biased due to negative weight problem (Borusyak, Jaravel, and Spiess 2024; Chaisemartin and D’Haultfoeuille 2020; Goodman-Bacon 2021) (see Roth et al. 2023, for elaboration). There are various estimators for different set of situations in a staggered adoption of DID. And the specific choice of estimator will depend on the type data, data quality and context.

I use Callaway and Sant’Anna (2021) (CS) estimator in this study to deal with treatment timing and treatment group heterogeneity to estimate the intention-to-treat (ITT) effects of health insurance eligibility. I do so firstly by aggregating the data at quarterly level. Y_{it} is the outcome for district i at time t . CS estimator estimates the ATT effects for group g , which is treated at time t . In my estimation g is defined as the cohort of units i (districts) that implement the treatment in the same period t (quarter-year). CS estimator works with a relatively more intuitive set of control groups, which in my case are those districts who have not implemented the health insurance by time t , known as "not yet treated" groups in the literature. CS estimator also imposes a weaker parallel trends assumption because it only relies on post treatment parallel trends, rather than imposing parallel trends for all the districts and time and is more efficient when the outcome might be serially correlated like in this scenario (Roth et al. 2023). CS estimator also provides an easy way of incorporating anticipation periods. And although I argue that the timing of treatment is quasi-random, CS estimator also accommodates the inclusion of covariates in case parallel trends assumption is only satisfied under inclusion of covariates.

Let G_i denote the first quarter in which district i becomes eligible for NHIP, with $G_i = \infty$ for districts not yet treated by the end of the sample. I refer to the set of distinct treatment cohorts as $\mathcal{G} = \{g : g \in \{9, \dots, \tau\}\}$, where the first cohort is treated in quarter 9. I set up the estimation using the potential outcomes framework. Let $Y_{it}(0)$ denote the potential outcome of district i at

time t if it remains untreated, and $Y_{it}(g)$ the potential outcome if first treated in quarter g . Define $\mathbb{1}\{G_i = g\}$ as the indicator for district i belonging to cohort g . The observed outcome can then be written as:

$$Y_{it} = Y_{it}(0) + \sum_{g \in \mathcal{G}} (Y_{it}(g) - Y_{it}(0)) \cdot \mathbb{1}\{G_i = g\} \quad (1)$$

The key estimand is the group-time average treatment effect on the treated:

$$ATT(g, t) = \mathbb{E}[Y_t(g) - Y_t(0) \mid G_i = g] \quad (2)$$

The CS estimator allows aggregation of group-time $ATT(g, t)$ estimates into summary measures. For the main analysis I use “group” aggregation as the measure of the overall ATT effect. For each cohort g , define the cohort-specific average effect as:

$$\theta(g) = \frac{1}{\tau - g + 1} \sum_{t=g}^{\tau} ATT(g, t) \quad (3)$$

The overall group aggregation then averages across cohorts, weighted by cohort size:

$$\theta^O = \sum_{g \in \mathcal{G}} \theta(g) \cdot P(G_i = g \mid G_i \leq \tau) \quad (4)$$

$\theta(g)$ is the average treatment effect of NHIP eligibility among all districts in cohort g across all post-treatment periods. Thus θ^O first calculates the average ATT for each cohort, then averages across cohorts to produce the summary measure.

I also estimate dynamic (event-study) ATT effects that show how the treatment effect evolves with time since adoption. Defining $e = t - g$ as the number of quarters since treatment, the dynamic ATT at event time e is:

$$\theta_{es}(e) = \sum_{g \in \mathcal{G}} \mathbb{1}\{g + e \leq \tau\} \cdot P(G_i = g \mid G_i + e \leq \tau) \cdot ATT(g, g + e) \quad (5)$$

$\theta_{es}(e)$ is the average effect across all cohorts observed exactly e quarters after treatment, weighted by the probability of belonging to cohort g conditional on being observed at event time e .

The validity of this strategy rests on several key assumptions. First, treatment is irreversible: once a district becomes eligible for NHIP, it remains eligible throughout the study period. This holds by design.

Second, Equation 2 provides a causal interpretation under the conditional parallel trends assumption with limited anticipation. Let $\delta \geq 0$ denote the anticipation period. For each cohort $g \in \mathcal{G}$ and for all pre-treatment periods t such that $t < g - \delta$:

$$\mathbb{E}[Y_t(0) - Y_{t-1}(0) \mid G_i = g] = \mathbb{E}[Y_t(0) - Y_{t-1}(0) \mid G_i \geq t + \delta] \quad (6)$$

Equation 6 states that, in the absence of treatment, the evolution of potential outcomes for cohort g districts would have been the same as for districts not yet treated by time $t + \delta$. The limited anticipation assumption allows districts to adjust their behaviour up to δ periods before treatment, but not earlier. In my context, the gap between registration opening and eligibility to claim benefits is approximately three months, so I set $\delta = 1$ quarter to account for this administrative delay. This means the parallel trends assumption is evaluated relative to the period before registration rather than before eligibility onset.

This assumption is likely to hold because the timing of NHIP implementation can be considered quasi-random: the order of rollout was not publicly known in advance, and actual implementation dates were influenced by administrative delays. To support this claim, I regress rollout timing on pre-treatment district characteristics — including baseline utilization levels, population, number of hospitals, and poverty rates — and find that none of the observables individually or jointly predict treatment timing (F-test p-value = 0.35; Table A2). The assumption cannot be tested directly, but I further assess its plausibility through three complementary approaches. First, the event-study plots in Figure 6 allow visual inspection of pre-treatment trends: the pre-treatment coefficients are close to zero and show no systematic pattern for all primary outcomes. Second, I conduct a joint Wald test of the null hypothesis that all pre-treatment dynamic ATT coefficients are simultaneously equal to zero. The test fails to reject the null for all primary outcomes (Table A3), providing formal evidence consistent with parallel pre-treatment trends. As Roth et al. (2023) emphasises, pre-trends tests have limited power against plausible violations of parallel trends, so a failure to reject should not be interpreted as definitive validation. Third, I include falsification

tests showing that NHIP has no effect on outcomes covered by a pre-existing free programme (the Safe Motherhood Program), which would not be expected to respond to insurance eligibility.

4.3. Inference

CS estimator can be estimated using outcome regression, inverse probability weighting (IPW) and doubly robust methods. Although the identification remains the same with all the methods, doubly robust method is more robust to misspecification compared to the other two (Sant'Anna and Zhao 2020). Therefore, in all my specification I use doubly robust method, with standard errors clustered at the district level with bootstrapping for 1000 iterations. All the confidence bands reported are simultaneous confidence bands, which are wider than the point wise confidence bands. I use them because they take into account the dependency across group-time average treatment effect and asymptotically cover the whole path of these effects (Callaway and Sant'Anna 2021). I use short gaps (varying base period) for the dynamic ATTs, which means that I estimate the treatment effect if the treatment had occurred in that period. However, choosing short gap over long differences (universal base period) does not affect my interpretation or the look of the event study plot, which is mainly because the post adoption ATTs are same regardless of the base period. As a robustness check, I re-estimate the main specifications using a universal base period and confirm that the results are unchanged (Table A7).

Since I test multiple utilization outcomes simultaneously, I address the concern of multiple hypothesis testing by applying the Romano and Wolf (2005) stepdown procedure, which controls the familywise error rate while accounting for the dependence structure across test statistics. I implement this using the influence functions from the CS estimator to construct the joint covariance matrix across all eight utilization outcomes. All outcomes remain statistically significant after the Romano-Wolf correction (Table A4).

4.4. Threats to Identification

The staggered rollout of NHIP between 2016 and 2022 coincides with three major events that could confound the estimated treatment effects if their impact on healthcare utilization varied systematically across districts in a manner correlated with NHIP timing. I discuss each in turn.

2015 Gorkha Earthquake. In April 2015, a magnitude-7.8 earthquake struck central Nepal, causing widespread destruction of health infrastructure in 14 of the 77 districts. The earthquake occurred approximately one year before the first NHIP pilot districts began implementation in 2016. If earthquake-affected districts were systematically earlier or later NHIP adopters, post-earthquake reconstruction and health system recovery could generate differential utilization trends that the estimator would incorrectly attribute to insurance. Two features of the setting mitigate this concern. First, the three pilot districts (Kailali, Baglung, and Ilam) are geographically distant from the earthquake epicentre and were not among the severely affected districts. Second, the CS estimator compares each treated cohort to not-yet-treated districts at each point in time, so any common recovery trends across earthquake-affected districts are differenced out provided they do not vary with NHIP timing within the affected group. As a robustness check, I re-estimate the main specifications excluding the 14 earthquake-affected districts and find that the results remain stable ([Table A5](#)).²

2017 Federalization Reform. Nepal's 2015 Constitution established a federal system of governance, and the transition to federalized health administration took effect in 2017. This reform restructured health service delivery by devolving responsibility from the central government to provincial and local governments, potentially disrupting administrative capacity and healthcare supply during the transition period. Crucially, federalization was a nationwide reform that affected all 77 districts simultaneously. In the CS framework, any common shock that affects all districts equally in a given time period is absorbed by the time fixed effects implicit in the group-time ATT estimation. Federalization threatens identification only if its disruptive effects varied across districts in a manner correlated with NHIP rollout timing. While I cannot fully rule this out, there is no institutional reason to expect that the federal transition differentially disrupted health services in districts that happened to adopt NHIP earlier versus later, as the two reforms were administered by different government agencies with independent implementation timelines.

COVID-19 Pandemic. The COVID-19 pandemic, beginning in March 2020, represents the most significant threat to identification. Nationwide lockdowns sharply reduced healthcare

2. The 14 severely affected districts are identified based on the National Planning Commission's Post-Disaster Needs Assessment (National Planning Commission [2015](#)).

utilization across all districts, and the pandemic’s impact likely varied across districts depending on infection rates, lockdown severity, and health system capacity. Since NHIP continued to roll out to new districts during 2020–2022, later treatment cohorts experienced the pandemic during both their pre-treatment and post-treatment periods, while earlier cohorts experienced it only in later post-treatment periods. If COVID-19’s impact on utilization was correlated with NHIP timing, the parallel trends assumption could be violated. To address this concern, I re-estimate the main specifications using only the pre-pandemic sample (through Q4 2019), which excludes the pandemic-affected periods entirely. The results from this truncated sample are consistent with the full-sample estimates, though with wider confidence intervals due to the reduced number of post-treatment periods for later cohorts (Table A6).

DHIS2 Reporting Composition. A potential concern with administrative data from the District Health Information System is that NHIP empanelment could change the composition or completeness of facility reporting, generating measured utilization increases that reflect improved data capture rather than genuine changes in health-seeking behaviour. If newly empanelled private facilities begin reporting to DHIS2 upon joining the insurance network, the district-level aggregates would mechanically increase even absent any change in actual utilization. Three features of the data mitigate this concern. First, DHIS2 reporting was mandated for all public health facilities from 2014, two years before the first NHIP pilot, establishing the reporting infrastructure independently of the insurance rollout. Second, the district-level data used in this analysis are compiled from a consistent set of facilities that reported throughout the 2016–2022 study period, constituting a balanced facility panel by construction. This ensures that the estimated effects capture changes in visits at the same facilities over time, not the entry of new reporting units. Third, the close alignment between DHIS2-implied utilization changes and independently observed insurance claims (Table 9) provides further reassurance that the measured increases reflect genuine insurance-financed visits rather than reporting artefacts. As an additional robustness check, I re-estimate the main specifications restricting the sample to public facilities only, which have the most consistent reporting history, and find that the results are qualitatively similar (Table A8).

Cross-District Spillovers (SUTVA). The validity of the staggered DiD design also requires that NHIP implementation in one district does not affect healthcare utilization in other districts. The institutional design of NHIP effectively rules out the most salient spillover channel. Registration is strictly residence-based: individuals can only enroll in the district where they reside. Upon enrollment, each member is assigned a first point of contact facility within their district for primary care. Crucially, even when a patient is referred to a facility in another district, the referral is not covered by the insurance program unless the patient is enrolled in that district. This means that residents of not-yet-treated districts cannot access insurance-financed care in treated districts, eliminating the possibility that cross-district patient flows inflate utilization counts in treated areas or deflate them in control areas. The remaining potential for spillovers—such as general equilibrium effects on provider supply or information diffusion about insurance across district borders—is likely to be small given the local nature of primary healthcare delivery in Nepal.

5 — Results

5.1. *Effects of health insurance on health visits*

I show the effects of eligibility of health insurance on hospital visits in [Table 2](#), Panel A. The outcome is aggregated at the quarterly periods for each district and I use group aggregate group time effect (AGGTE) from the CS estimator for the analysis of overall treatment effect.

I find that eligibility of health insurance increases the total visits, including all inpatient, outpatient, and emergency services, to health service providers by 14.2 percent ($e^{0.133} - 1$).³ The results for outpatient services follow a similar pattern. Total outpatient department (OPD) visits increase by 13.7 percent ($e^{0.128} - 1$), while new OPD visits rise by 12.4 percent ($e^{0.117} - 1$). The largest effects appear in emergency care where the total emergency visits increase by 19.6 percent ($e^{0.179} - 1$), and new emergency visits by 18.9 percent ($e^{0.173} - 1$). This pattern is consistent with NHIP’s policy design, which exempts emergency admissions from the referral requirement at the first point of contact. Patients may therefore be using emergency services as an alternative pathway to access care without navigating the referral system. Appendix [Table A1](#) reports these

3. Throughout this paper, I convert log-linear coefficients to percentage effects using the exact transformation $(e^{\hat{\beta}} - 1) \times 100$, which avoids the approximation error of $\hat{\beta} \times 100$ that grows with the magnitude of $\hat{\beta}$.

effects in per-capita terms (visits per 1,000 population) to facilitate comparison of magnitudes across districts of different sizes.

Table 2 — Effects of Health Insurance Eligibility on Healthcare Utilization

	Estimate (Is NHIP eligible = 1)	Std. Error	Pre-treatment Mean
Panel A: Overall Healthcare Visits			
Total Visits	0.1331***	(0.0234)	101,979
OPD Visits	0.1281***	(0.0307)	78,356
Emergency Visits	0.1790***	(0.0646)	9,275
Total Visits (Male)	0.1426***	(0.0261)	42,852
Total Visits (Female)	0.1266***	(0.0231)	59,127
Female Share of Visits	-0.0038**	(0.0019)	0.5806
Panel B: Number of OPD Visits by Disease Category			
OPD Communicable	0.1472***	(0.0269)	8,286
OPD Non-Communicable	0.1363***	(0.0445)	36,829
OPD Injuries	0.1302***	(0.0397)	13,322
OPD CVD	0.0072	(0.1218)	848
OPD Cancer	-0.0064	(0.0439)	937
Panel C: Diagnostic Services			
Laboratory Services	0.9580***	(0.3569)	11,569
Radiographic Services	0.1703**	(0.0698)	20,093
<i>Fit statistics</i>			
Number of Cohorts	17		
Number of Time Periods	30		

Note: All outcomes are log-transformed. Panel A shows effects on total visits across all services, OPD, and emergency care. Panel B disaggregates OPD visits by disease category. Panel C shows effects on diagnostic services. Pre-treatment mean shows the average outcome value before NHIP implementation. Clustered (District) standard errors in parentheses, bootstrapped with 1000 iterations. Estimation Method: Doubly Robust. Control Group: Not Yet Treated. Anticipation Periods: 1. Signif. Codes: *** 99% ** 95% * 90% uniform confidence band does not include 0.

Panel B of [Table 2](#) presents the effects of health insurance eligibility on OPD visits by disease category. Health insurance increases OPD visits for communicable diseases by 15.9 percent ($e^{0.147} - 1$), non-communicable diseases by 14.6 percent ($e^{0.136} - 1$), and injuries and surgical cases by 13.9 percent ($e^{0.130} - 1$). These estimates are comparable to the 13.7 percent increase in total OPD visits, suggesting that insurance does not disproportionately affect any particular condition type. However, health insurance eligibility has no significant effect on OPD visits for cardiovascular diseases or malignancies. This finding is consistent with existing policy: the Ministry of Health and Population operates a separate program for essential non-communicable diseases, including cardiovascular conditions and cancers, which provided free treatment prior to the introduction of health insurance. This also acts as a falsification test, showing that health insurance does not affect outcomes unrelated to its coverage.

Panel C of [Table 2](#) presents the effects of health insurance eligibility on the use of diagnostic services. Health insurance increases the use of laboratory services by 160.7 percent ($e^{0.958} - 1$) and radiographic services by 18.6 percent ($e^{0.170} - 1$). Since NHIP covers nearly all laboratory and radiographic services, these increases are consistent with the observed rise in OPD visits: as more patients seek outpatient care, demand for diagnostic services increases accordingly. The more-than-doubling of laboratory service use is particularly notable, reflecting substantial improvements in access to diagnostic care.

[Table 3](#) examines heterogeneity in the effects of health insurance eligibility across age groups. Health insurance increases health service visits across all age categories, with the largest effects observed among those aged 60 and above (19.9 percent, $e^{0.181} - 1$). The effects for the 10–19 and 20–59 age groups are similar in magnitude, at 15.7 percent and 15.3 percent respectively. The smallest effect appears among children aged 0–9, at 9.6 percent. A similar pattern emerges for new visits. These findings are consistent with the well-documented positive relationship between age and healthcare demand.

Table 3 — Effects of Health Insurance Eligibility on Health Visits by Age Group

	Estimate (Is NHIP eligible = 1)	Std. Error	Pre-treatment Mean
Age 0-9	0.0917***	(0.0245)	18,272
Age 10-19	0.1461***	(0.0292)	17,143
Age 20-59	0.1420***	(0.0241)	52,668
Age 60+	0.1812***	(0.0296)	13,896
<i>Fit statistics</i>			
Number of Cohorts	17		
Number of Time Periods	30		

Note: All the outcomes are log transformed. Clustered (District) standard-errors in parentheses bootstrapped with 1000 iterations. Estimation Method: Doubly Robust. Control Group: Not Yet Treated, Anticipation Periods: 1. Signif. Codes: *** 99% ** 95% * 90% uniform confidence band does not include 0.

5.2. Poverty Heterogeneity

Another important aspect of health insurance in lower-middle income countries is that it is often targeted to reduce poverty. In [Table 4](#), I show the heterogeneous effect of health insurance on health visits and OPD morbidity by poverty incidence. Using pre-treatment poverty incidence rate of 2011 by World Bank, I group districts into low-, medium-, and high-poverty categories corresponding to the bottom quartile, interquartile range, and top quartile of the distribution. I find that the effects of health insurance on health visits are positive and statistically significant for richer districts, but statistically insignificant for poor districts across all kinds of visits. And similar results are found for OPD morbidity as well. The overall effects of health insurance therefore seem to be driven by rich districts.

Table 4 – Effects of Health Insurance Eligibility by District Poverty Level

	High poverty (Q4)		Middle poverty (Q2–Q3)		Low poverty (Q1)	
	Estimate	Std. Error	Estimate	Std. Error	Estimate	Std. Error
Panel A: Overall Healthcare Visits						
Total Visits	0.0256	(0.0456)	0.0557	(0.0311)	0.2039***	(0.0441)
OPD Visits	0.0621	(0.0545)	0.0008	(0.0563)	0.2018***	(0.0422)
Emergency Visits	0.0082	(0.0986)	0.1390	(0.1025)	0.2435***	(0.0874)
Panel B: OPD Visits by Disease Category						
OPD Communicable	0.1690***	(0.0392)	0.0407	(0.0562)	0.1946***	(0.0325)
OPD Non-Communicable	0.0472	(0.0416)	-0.0058	(0.0654)	0.1994***	(0.0477)
OPD Injuries	0.0401	(0.0476)	0.0071	(0.0478)	0.2214***	(0.0543)
<i>Fit statistics</i>						
Number of Cohorts	9		14		12	
Number of Time Periods	27		27		30	

Note: This table presents heterogeneous effects of health insurance eligibility by district poverty level. All outcomes are log-transformed. Panel A shows effects on overall visits. Panel B disaggregates OPD visits by disease category. The stark contrast between rich and poor districts highlights substantial inequalities in program benefits. Clustered (District) standard errors in parentheses, bootstrapped with 1000 iterations. Estimation Method: Doubly Robust. Control Group: Not Yet Treated. Anticipation Periods: 1. Signif. Codes: *** 99% ** 95% * 90% uniform confidence band does not include 0.

5.2.1 Mechanisms

The null effects in high-poverty districts could reflect either demand-side barriers—such as enrollment frictions or low willingness to seek care—or supply-side constraints limiting access to covered services. To investigate these mechanisms, I examine enrollment patterns and healthcare infrastructure across these three groups in [Table 5](#). I use individual level registration and claims data from HIB to compute district-level averages for enrollment rates and utilization among enrollees. I also compile data on the number of hospitals and NHIP-empanelled providers in each district from HIB records.

Table 5 — Mechanisms Behind Poverty Heterogeneity

	High Poverty	Medium Poverty	Low Poverty
Panel A: Demandside			
Registration Rate (per 1,000 pop.)	20.0	23.3	27.9
Share Claimed (conditional on enrollment)	0.191	0.302	0.383
Panel B: Supplyside			
Number of Registered Health Facilities	92	114	132
Number of Empanelled Facilities	3.3	6.2	11.6

Note: Authors own calculation using Registration and Claims data. Registration rate is the average number of individuals enrolled in NHIP per 1,000 district population. Any Claim measures the proportion of enrollees who made at least one insurance claim during the post-treatment period. Number of health facilities includes all public and private healthcare providers in the district. Number of empanelled facilities refers to healthcare providers contracted to deliver NHIP-covered services. All figures are district-level averages within each poverty distribution.

Panel A of [Table 5](#) shows that enrollment patterns differ substantially across poverty groups. Individual-level registration rates are 28 percent lower in high-poverty districts compared to low-poverty districts (20.0 versus 27.9 per 1,000 population), suggesting that demand-side barriers—whether premium costs, documentation requirements, or limited program awareness—constrain take-up in poorer areas. More strikingly, even among those who do enroll, utilization rates diverge sharply: only 19 percent of enrollees in high-poverty districts make at least one insurance claim, compared to 38 percent in low-poverty districts. This pattern indicates that enrollment alone does not translate into equivalent healthcare access. This is similar to what is experienced in Indonesia (Banerjee et al. [2021](#)).

Panel B reveals a potential explanation for the utilization gap conditional on enrollment. High-poverty districts have substantially fewer healthcare facilities overall (92 versus 132), but the disparity in NHIP-empanelled facilities is particularly stark: poor districts average only 3.3 empanelled providers compared to 11.6 in richer districts—a 3.5-fold difference. With so few contracted providers, enrollees in poor districts face limited options for accessing covered care.

They may need to travel longer distances to reach an empanelled facility, or forgo insurance benefits entirely and seek care from non-empanelled providers at full cost.

These findings suggest that the null utilization effects in high-poverty districts reflect a combination of enrollment frictions and supply-side constraints. Expanding insurance coverage alone appears insufficient to improve healthcare access in underserved areas; complementary investments in healthcare infrastructure and provider empanelment may be necessary to realize the program's equity objectives.

5.2.2 Within-District Poverty Gradient

To distinguish demand-side barriers from district-level supply constraints, I exploit within-district variation in municipality-level poverty. Using Small Area Estimation poverty rates at the ilaka level from the 2011 Census, mapped to current municipality boundaries via a federal restructuring concordance (749 municipalities matched, covering 100% of registrants), I construct a municipality $\times$ quarter panel and progressively add controls to decompose the poverty gradient.

District $\times$ quarter fixed effects absorb all district-level time-varying confounders—including the number and type of empanelled providers, NHIP rollout timing, and district-level health infrastructure. The identifying variation is thus across municipalities within the same district-quarter. I then progressively add geography controls (elevation, terrain ruggedness, road access, land area), demand-side proxies (rural status, education), and a supply-side control (log number of unique assigned First Point of Contact facilities per municipality) to decompose the sources of the poverty gradient.

Table 6 — Within-District Poverty Gradient: Registration and Claims

	Reg. per 1,000 pop				Claims per Registrant			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Municipality Poverty (FGT0)	-676.3*** (99.47)	-578.6*** (99.63)	-355.4*** (98.02)	-165.7* (88.45)	-0.209*** (0.031)	-0.171*** (0.037)	-0.110*** (0.034)	-0.069* (0.039)
District × Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Geography controls	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Urbanization + Education	No	No	Yes	Yes	No	No	Yes	Yes
Supply control	No	No	No	Yes	No	No	No	Yes
Observations	13,122	13,122	13,122	13,122	13,122	13,122	13,122	13,122
Unique municipalities	748	748	748	748	748	748	748	748
R^2	0.762	0.764	0.776	0.786	0.508	0.513	0.521	0.524

Notes: The unit of observation is a municipality×quarter, restricted to municipality-quarters with positive registrations. Municipality poverty (FGT0) is the population-weighted ilaka-level headcount rate from Small Area Estimation (CBS/World Bank, 2011 Census), mapped to current municipality boundaries via a federal restructuring concordance. Geography controls: log elevation, log ruggedness, log land area, road access. Urbanization + Education: rural indicator, share with intermediate education or above. Supply control: log number of unique assigned First Point of Contact (FPOC) facilities per municipality—an exogenous measure reflecting designated provider availability, not utilization choices. Clustered (municipality) standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 6 presents the results. Column (1) shows the raw within-district poverty gradient: a 10 percentage point increase in municipality poverty is associated with 68 fewer registrations per 1,000 population and 2.1 fewer claims per 100 registrants. Adding geography controls (column 2) attenuates the registration coefficient by 14%, suggesting remoteness plays a modest role. Adding urbanization and education (column 3) produces further attenuation, with the registration coefficient falling to -355—rural municipalities register significantly less, and education is positively associated with both outcomes, consistent with demand-side barriers related to information and health literacy.

The critical test is column (4), which adds the number of assigned healthcare facilities (unique FPOCs) as a within-district supply control. This measure is exogenous—facilities are designated at

registration, not chosen by enrollees. Adding supply attenuates the poverty coefficient further but it remains negative ($-165.7, p < 0.10$). The decomposition suggests that geography explains roughly 14% of the within-district poverty gradient, demand-side factors (urbanization and education) explain 33%, facility supply explains 28%, and a residual 25% reflects poverty-related demand barriers beyond observable characteristics.

Figure 4 visualizes the unconditional relationship between municipality poverty and insurance outcomes.

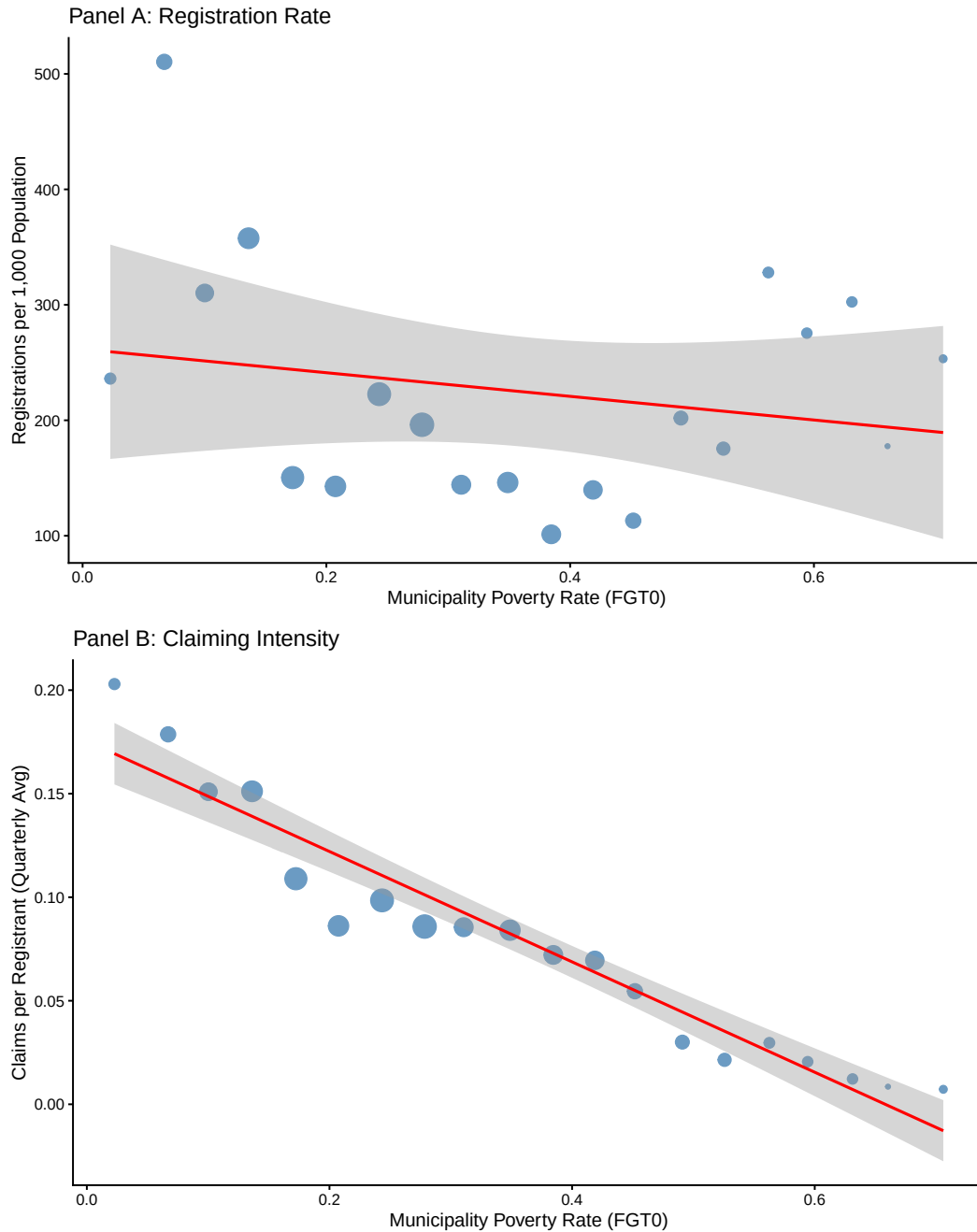


Figure 4 – Municipality Poverty and Insurance Outcomes

5.2.3 Distance as a Demand-Side Barrier

To provide causal evidence on the role of access barriers, I exploit the 2019 policy change that restructured First Point of Contact (FPOC) designation, prioritizing public hospitals over private facilities. Prior to this reform, enrollees could choose any empanelled facility—public or

private—as their FPOC. The 2019 reform reshuffled FPOC assignments across the system, though implementation was not uniform: while the policy directed enrollees toward public facilities, a substantial share of treated individuals retained or were assigned to private FPOCs, reflecting incomplete enforcement and the continued role of NGO-run facilities classified here as private.⁴ Using individual-level claims data with road distances computed via Google Maps and OSRM routing, I examine whether this distance shock operated differentially across poverty groups. The sample is restricted to individuals observed claiming in both the pre- and post-2019 periods, which means results apply to the intensive margin of active claimants and represent a selected subsample of all enrollees.⁵

Table 7 presents the effect of the FPOC policy change on distance to care, estimated with individual fixed effects and ward×post fixed effects, with ward-clustered standard errors following Dupas and Jain (2024). The ward×post interactions absorb ward-level time shocks—including differential COVID-19 exposure across wards—ensuring that the treatment effect is identified from within-ward variation in FPOC reassignment. The key coefficient—the interaction of treatment (FPOC reassigned) with the post-2019 indicator—captures the differential distance change for individuals whose designated facility was altered by the policy. In the full sample, reassignment reduced distance by approximately 0.85 km on average, with similar magnitudes across poverty groups: −0.75 km in high-poverty districts ($p < 0.05$) and −0.87 km in low-poverty districts ($p < 0.01$). The similarity of the first-stage effect across poverty groups (interaction $p = 0.77$) indicates that the policy generated a comparable distance shock regardless of local infrastructure, providing a clean setting to test whether distance reductions translate differently into utilization.

4. Of the 501 hospitals in the claims data, 426 (85%) are public and 75 (15%) are private. NGO-run facilities (e.g., Nepal Netra Jyoti Sangh eye hospitals, mission hospitals) are classified as private following local convention. Among treated individuals, 39% have a private post-2019 FPOC, reflecting both the gradual nature of implementation and the classification of NGO facilities.

5. This sample restriction means that individuals who stopped claiming after the policy change—potentially those most affected by increased distance—are excluded. However, a formal attrition test (Appendix Table A19) shows that among pre-2019 claimants with a recorded FPOC, those with private FPOCs (more likely affected by the policy) had lower attrition (30.6%) than those with public FPOCs (33.2%), ruling out differential attrition as a source of bias.

Table 7 — Effect of 2019 FPOC Policy on Distance to Care

	Full Sample	High Poverty	Low Poverty
	(1)	(2)	(3)
Treated $\times$ Post	-0.854***	-0.746**	-0.868***
	(0.182)	(0.375)	(0.200)
<i>Fixed effects</i>			
Individual FE	Yes	Yes	Yes
Ward $\times$ Post	Yes	Yes	Yes
<i>Summary statistics</i>			
Mean distance (km)	7.6	7.4	7.7
Unique FPOC hospitals	289	96	200
Observations	697,698	147,000	550,698
N (individuals)	373,281	80,478	292,803
N (wards)	1,855	827	1,028

Note: This table reports the effect of the 2019 FPOC policy change on road distance (km) to the designated First Point of Contact hospital. Treated $\times$ Post captures the differential distance change for individuals whose FPOC was reassigned following the restriction to public hospitals. All specifications include individual fixed effects and ward $\times$ post interactions to absorb ward-level time shocks, with ward-clustered standard errors. Poverty groups defined by median split of district poverty rate. Distance outliers above 25km excluded. Signif. Codes: *** 1% ** 5% * 10%.

Note: Dependent variable is road distance (km) from ward centroid to FPOC hospital. All specifications include individual fixed effects and ward $\times$ post interactions to absorb ward-level time shocks. Ward-clustered standard errors in parentheses following Dupas and Jain (2024). Distance outliers above 25 km excluded. Appendix Table A12 reports robustness to alternative fixed effects; Table A13 tests sensitivity to COVID exposure; Table A14 varies the distance bandwidth. Signif. Codes: *** 1% ** 5% * 10%.

Table 8 directly estimates the reduced-form effect of FPOC reassignment on insurance claims. This approach asks: did the policy-induced disruption in provider assignment affect utilization differently across poverty groups? The results show that FPOC reassignment increased claims by 0.64 on average, with the effect concentrated in low-poverty districts (0.69) and substantially smaller—though still statistically significant—in high-poverty districts (0.23). Since the first-stage distance shock is similar across poverty groups, the differential reduced-form response implies that

the same distance reduction translates into much larger utilization gains in low-poverty districts. This pattern is consistent with a demand-side interpretation: enrollees in better-served areas are more responsive to distance reductions, while those in high-poverty districts face additional barriers—limited provider availability, lower awareness, or higher non-monetary costs—that attenuate the utilization response even when physical distance to care decreases.

Table 8 — Effect of FPOC Policy on Insurance Claims

	Full Sample	High Poverty	Low Poverty
	(1)	(2)	(3)
Treated $\times$ Post	0.636***	0.234***	0.692***
	(0.042)	(0.054)	(0.046)
<i>Fixed effects</i>			
Individual FE	Yes	Yes	Yes
Ward $\times$ Post	Yes	Yes	Yes
<i>Summary statistics</i>			
Mean claims	4.04	3.63	4.15
Unique FPOC hospitals	289	96	200
Observations	697,698	147,000	550,698
N (individuals)	373,281	80,478	292,803
N (wards)	1,855	827	1,028

Note: This table reports the reduced-form effect of the 2019 FPOC policy change on the number of insurance claims per individual. Treated $\times$ Post captures the differential change in claiming for individuals whose FPOC was reassigned following the restriction to public hospitals. All specifications include individual fixed effects and ward $\times$ post interactions to absorb ward-level time shocks, with ward-clustered standard errors. Poverty groups defined by median split of district poverty rate. Distance outliers above 25km excluded. Signif. Codes: *** 1% ** 5% * 10%.

Figure 5 visualizes the distribution of distance changes across poverty groups. The distributions are similar in shape, confirming that the policy generated a comparable distance shock across poverty groups. The key finding is not that the shock differed, but that the same shock produced markedly different utilization responses.

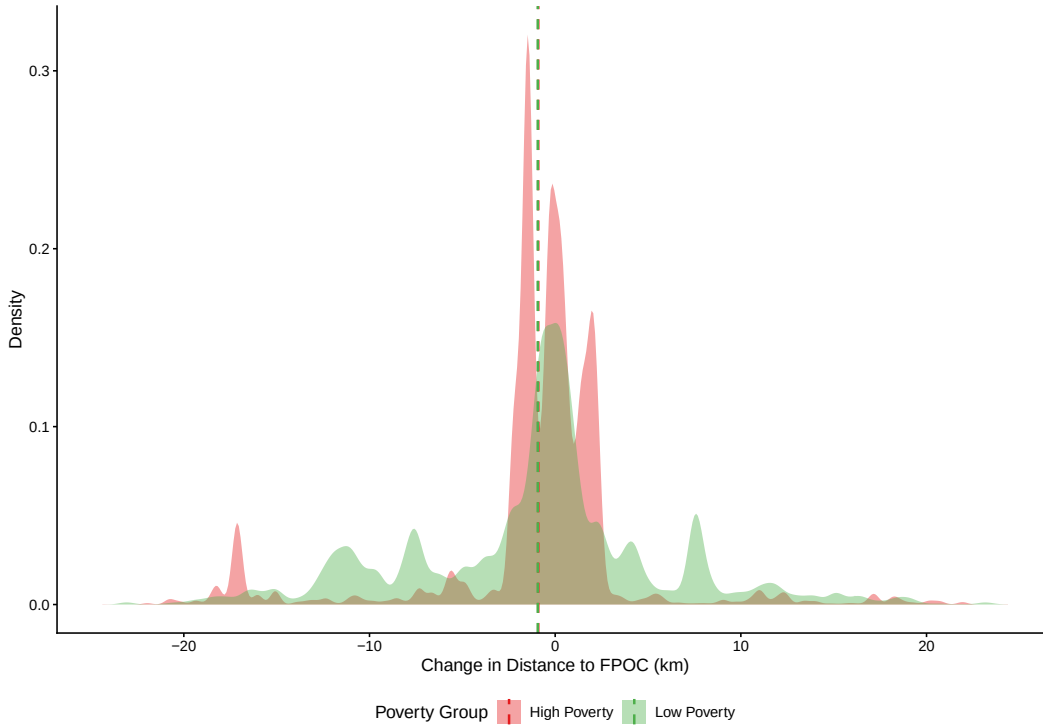


Figure 5 — Distribution of Distance Changes to FPOC by Poverty Group

The sensitivity of the distance-utilization relationship to specification deserves comment. Instrumental variable estimates—where FPOC reassignment instruments for distance to care—are reported in Appendix [subsection A.6](#). The preferred ward $\times$ post specification implies that each additional kilometer of distance reduces claims by approximately 0.7 per individual-period. However, the IV should be interpreted cautiously. The FPOC policy shifted distance in opposite directions for different transition types—reducing it for public-to-public reassignments but increasing it for public-to-private ones—which violates the monotonicity assumption required for a well-defined LATE. The Appendix provides a detailed examination of transition-type heterogeneity, the exclusion restriction, and hospital quality proxies that support the interpretation that distance, rather than provider type or quality, is the primary channel. Appendix [Table A12](#) further shows that the reduced-form results require individual fixed effects: coarser fixed effects (VDC, ward) produce sign reversals because treated and untreated individuals differ systematically in baseline characteristics. The results are robust to replacing ward $\times$ post with VDC $\times$ post interactions, to restricting the sample to enrollment cohorts with minimal COVID overlap (Appendix [Table A13](#)), and to varying the distance bandwidth between 10 and 40 km (Appendix [Table A14](#)).

To further assess the plausibility of my estimates, [Table 9](#) compares the magnitude and composition of utilization changes implied by the difference-in-differences results with observed insurance claims from an independent administrative source. Using the DiD coefficients and pre-treatment means from the DHIS2 facility records, I compute the implied increase in total quarterly visits per district using the exact transformation $(e^{\hat{\beta}} - 1) \times \bar{Y}_{pre}$ and contrast these figures with the average number of insurance-financed visits observed in claims data for treated districts. Despite differences in coverage and measurement across the two data sources, the magnitudes are closely aligned: the DiD estimates imply an increase of approximately 14,500 visits per district–quarter, while claims data record an average of about 10,900 insured visits over the same period. Similar consistency is observed across gender and age groups, with both data sources indicating larger utilization increases among working-age adults. The poverty breakdown in Panel C warrants a caveat: while claims data record insurance-financed visits in all three poverty groups, the DiD coefficient for high-poverty districts is not statistically significant ([Table 4](#)), so the implied figure of 2,644 visits is imprecisely estimated. The claims figure of 2,246 visits in high-poverty districts reflects actual insurance use by the small number of enrollees in those areas, rather than a precisely estimated causal effect of NHIP eligibility on overall utilization. This is consistent with the mechanisms discussed above: low enrollment and limited provider empanelment in poor districts constrain the program’s aggregate impact even when individual enrollees do access care. Taken together, this comparison suggests that the increase in healthcare utilization documented in the DiD analysis is largely driven by insurance-financed care, and that measurement differences across administrative systems are unlikely to drive the main results.

Table 9 – Comparison of DiD-Implied Utilization Changes with Insurance Claims

	Claims Data (Observed)	DHIS2 Data (Implied Effect)	Table Reference
Panel A: Overall Visits and Gender Composition			
Total Visits	10,876	14,518	Table 2
Total Male Visits	4,549	6,568	Table 2
Total Female Visits	6,327	7,980	Table 2
Female Share of Visits	0.5597	0.5768	Table 2
Panel B: Age Composition			
Age 0–9	707	1,755	Table 3
Age 10–19	724	2,697	Table 3
Age 20–59	6,409	8,036	Table 3
Age 60+	2,865	2,761	Table 3
Panel C: Poverty Heterogeneity			
Total Visits (High Poverty)	2,246	2,644	Table 4
Total Visits (Middle Poverty)	7,129	5,841	Table 4
Total Visits (Low Poverty)	23,303	23,065	Table 4

Note: Claims data report the quarterly average number of insurance-financed visits per district among enrolled individuals in treated districts during the post-treatment period. DHIS2 figures are derived by applying the estimated DiD coefficients to pre-treatment mean utilization levels and represent implied changes in total facility visits. While the two data sources measure different populations and outcomes, the similarity in magnitudes and patterns across subgroups suggests that the increase in utilization documented in the DiD analysis is largely driven by insurance-financed care.

5.3. Out-of-Pocket Expenditures and Financial Protection

A central objective of public health insurance programs is to reduce household out-of-pocket medical expenditures (OOPE) and protect families from catastrophic health costs. However, recent evidence on the effectiveness of health insurance expansions in low- and middle-income countries shows mixed results. Dupas and Jain in Hanna and Olken (2026) review ten rigorous studies and find that only two of seven report significant reductions in OOPE, while the remainder find null

effects. Importantly, interpreting null OOPE results is complicated: if insurance reduces the unit price of care while simultaneously increasing utilization, households may spend similar amounts but receive substantially more services—an outcome that still represents improved welfare through better health access.

In this context, I examine whether NHIP eligibility affects household expenditure patterns using data from the Household Risk and Vulnerability Survey (HRVS) 2016–18. Table 10 presents the results. Despite the large increases in healthcare utilization documented in previous sections, I find no statistically significant effects on out-of-pocket health spending, total consumption, or food and non-food expenditures.

Table 10 — Health Insurance Eligibility on Expenditure

	OOPE Health	Total Consumption	Food Consumption	Non-Food Consumption	Education Expenditure
Model:	(1)	(2)	(3)	(4)	(5)
<i>Group Aggregation</i>					
Is NHIP eligible = 1	-0.4800 (0.4110)	-0.0389 (0.0673)	0.0256 (0.0564)	-0.1028 (0.1150)	0.0861 (0.1935)
<i>Fit statistics</i>					
Number of Cohorts	2	2	2	2	2
Number of Time Periods	3	3	3	3	3

Note: Data comes from HRVS 2016-18. The HRVS is a three-year panel survey covering a subset of districts. Clustered (District) standard-errors in parentheses bootstrapped with 1000 iterations.

Estimation Method: Doubly Robust. Control Group: Not Yet Treated, Anticipation Periods: 0. Signif.

Codes: *** 99% ** 95% * 90% uniform confidence band does not include 0.

These estimates are imprecise due to limited survey coverage—HRVS covers substantially fewer districts and survey waves compared to the facility-level DHIS2 panel—reducing statistical power relative to the utilization analysis. Moreover, the survey timing may not align well with the NHIP rollout in many districts, and household expenditure data are typically noisier than administrative records. These null results should therefore be interpreted cautiously and do not necessarily imply the absence of true effects.

Nevertheless, the pattern of results—large utilization increases with no detectable change in household spending—is consistent with two complementary interpretations supported by the validation exercise in [Table 9](#). First, the increased utilization operates primarily through insurance-financed care rather than increased out-of-pocket spending, as evidenced by the close alignment between DiD-implied utilization changes and observed insurance claims. Second, as Conti and Gupta (2023) note, if insurance lowers the effective price per visit while raising the quantity of care consumed, total household expenditure may remain stable even as families obtain more healthcare services. This interpretation is consistent with NHIP’s design as a comprehensive, low-premium insurance scheme that substantially reduces marginal costs of care for enrolled households.

The absence of measurable OOOPE reductions, however, does not necessarily imply the program failed to improve financial protection. The household survey samples both enrolled and non-enrolled individuals within treated districts, limiting the scope for detecting expenditure effects when enrollment rates are modest. Additionally, financial protection effects may materialize over longer time horizons as households adjust consumption and savings behavior in response to reduced health expenditure risk—dynamics not fully captured within the survey window. Future research with longer panel data and information on individual enrollment status would be valuable for assessing the financial protection impacts of NHIP more conclusively.

5.4. Dynamic ATT

The static estimates in the preceding sections summarise the overall effect of NHIP eligibility, but they mask how these effects evolve over time. [Figure 6](#) plots dynamic ATTs from the CS estimator, tracing the treatment effect at each quarter relative to implementation. I use a varying base period specification, so each estimate captures the change relative to the period immediately before it, rather than a single distant reference period.

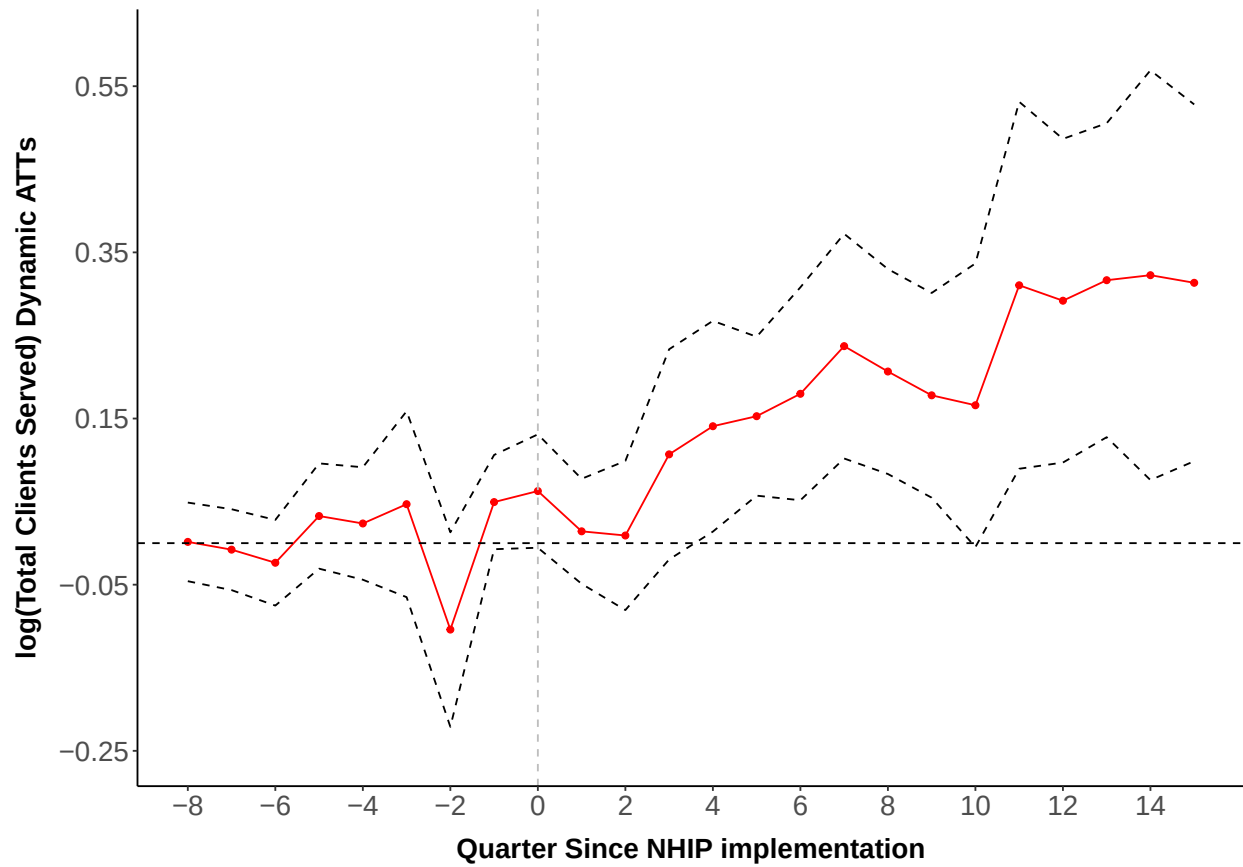
Evidence of causal validity comes from the pre-treatment coefficients on the left side of the figure. These hover around zero, rarely exceeding 0.05 in absolute value, and are statistically indistinguishable from zero throughout the pre-treatment window. Healthcare utilization in districts about to receive NHIP does not appear to have been diverging from utilization in not-yet-treated districts ahead of implementation. In other words, the estimated leads are consistent with the parallel trends assumption.

The post-treatment dynamics tell a clear story. In the first few quarters after implementation, the treatment effects are small and not significantly different from zero. This is unsurprising: NHIP coverage takes time to build. Registration must be processed, insurance cards distributed, and households must become aware of their benefits before utilization responds. The effects then grow steadily, becoming statistically significant approximately five quarters after implementation. By quarters 12–14, the point estimates reach approximately 0.35 log points, corresponding to a 42 percent increase in total visits.

The effects appear to plateau beyond this horizon rather than continuing to grow, suggesting that the program reaches a coverage ceiling as the pool of willing enrollees in each district saturates. The widening confidence intervals at longer horizons reflect a mechanical feature of staggered adoption designs: only the earliest-treated cohorts contribute to long-horizon estimates, so the effective sample shrinks as event time increases. This does not indicate that the effects themselves are becoming less precise in any economic sense—it simply means that fewer districts inform these estimates.

Dynamic ATTs for other outcomes exhibit a similar pattern: near-zero pre-treatment leads, gradual post-treatment emergence, and eventual levelling off. This consistency across outcomes reinforces the interpretation that the measured effects reflect genuine changes in healthcare utilization driven by insurance coverage, rather than artefacts of a particular outcome measure or specification choice.

Figure 6 — Total Clients Served Dynamic ATTs



Notes. This figure shows dynamic ATTs for total clients served by health service providers. The outcome is log-transformed total clients served. The x-axis shows quarters since NHIP implementation, with quarter 0 being the first implementation quarter. The shaded area represents a 95% simultaneous confidence band, constructed using the bootstrap procedure in Callaway and Sant’Anna (2021).

6 — Falsification Exercise

Although NHIP funded for pregnancy-related delivery services, I do not expect NHIP to have any substantial effect on outcomes related to maternal mortality, still births, institutional delivery among others. This is because of Safe Motherhood Program that was launched in 2009 to reduce maternal/neonatal morbidity and mortality, increase institutional delivery, increase the share of births attended by skilled professionals and other factors related to safe pregnancy. It was implemented nationwide at the community level providing incentives for institutional delivery via free transportation, incentivizing antenatal care and providing institutional delivery services

for free. It was eligible in any public health facility with a birthing facility for any mother and newborn who are a Nepali citizen. In this sense, the Safe Motherhood Program eliminated any financial barriers that NHIP would have otherwise addressed in accessing this kind of services. I show the results of the falsification exercise in [Table 11](#). Consistent with the expectations, I show that health insurance has no effect on outcomes related to the safe motherhood program across range of different outcomes. I show the effects with various simple, group and calendar aggregation and all the estimates cannot be distinguished from zero.

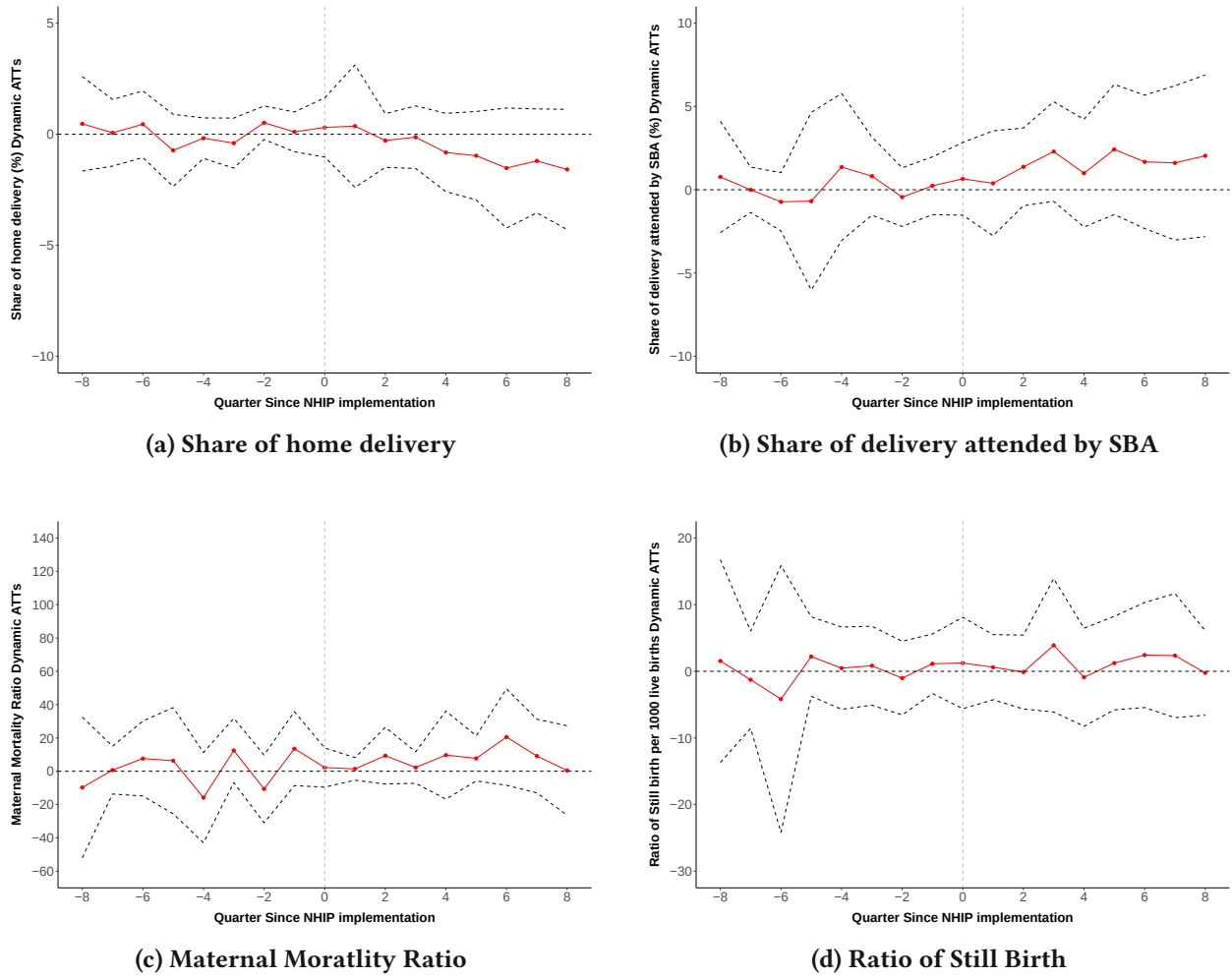
Table 11 – Health Insurance Eligibility on Safe Motherhood

	Share of home delivery	Share of delivery attended by SBA	Maternal Mortality Ratio	Ratio of still birth (per 1000 live births)
Model:	(1)	(2)	(3)	(4)
<i>Simple Aggregation</i>				
Is NHIP eligible = 1	-1.0325 (0.7063)	1.5733 (1.0995)	12.4370 (7.6634)	0.6529 (1.6171)
<i>Group Aggregation</i>				
Is NHIP eligible = 1	-0.3403 (0.7386)	1.5517 (0.8783)	6.8191 (6.2901)	1.6647 (1.7483)
<i>Calendar Aggregation</i>				
Is NHIP eligible = 1	-0.5074 (0.6344)	0.9444 (0.9356)	7.8861 (5.4218)	0.4202 (1.4762)
Pre-Treatment Mean	7.16	89.17	12.27	17.11
<i>Fit statistics</i>				
Number of Cohorts	17	17	17	17
Number of Time Periods	30	30	30	30

Note: Clustered (District) standard-errors in parentheses bootstrapped with 1000 iterations. Estimation Method: Doubly Robust. Control Group: Not Yet Treated, Anticipation Periods: 1. Signif. Codes: *** 99% ** 95% * 90% uniform confidence band does not include 0.

I show the dynamic ATTs in [Figure 7](#). The dynamic ATTs further show that the effect on these variables cannot be distinguished from zero throughout the study period.

Figure 7 – Dynamic ATTs for Safe Motherhood outcomes



Note: Panel (a) shows the dynamic ATTs for share of home delivery. Panel (b) shows dynamic ATTs for share of delivery attended by skilled birth attended. Panel (c) shows dynamic ATTs for maternal mortality ratio. Panel (d) shows dynamic ATTs for ratio of still births per 1000 live births. These dynamic ATTs are from CS model, where x-axis is quarter since NHIP implementation and y-axis is outcome. The black dotted lines represent a 95% uniform confidence interval.

This null result supports that the main results are not driven by broad improvements in healthcare services or district-specific trends, making the causal interpretation of the main results credible.

7 – Conclusion

This paper provides new evidence on the effects of Nepal’s National Health Insurance Program using a staggered difference-in-differences design. Health insurance eligibility increases healthcare utilization substantially: total visits rise by 14.2 percent, outpatient visits by 13.7 percent, and emergency visits by 19.6 percent. The more-than-doubling of laboratory service use (160.7 percent) suggests meaningful improvements in access to diagnostic care. Dynamic effects reveal a gradual uptake process, with significant impacts emerging approximately five quarters after implementation.

However, the benefits of NHIP are not equitably distributed. Utilization gains are concentrated in low-poverty districts, with no statistically significant effects in high-poverty areas. Gender effects are modest: both male and female visits increase substantially in absolute terms, though the female share of total visits declines slightly, suggesting that men may have had greater unmet demand prior to the program—a pattern consistent with the gender disparities in insurance utilization documented by Dupas and Jain (2024) in India. These patterns raise concerns about the program’s effectiveness in reaching its most vulnerable target populations.

These findings contribute to the growing literature on health insurance in developing economies by highlighting a tension between aggregate utilization gains and distributional equity. The results underscore the importance of complementary interventions—such as targeted outreach, subsidies for poor households, and efforts to address gender-specific barriers—to ensure that public health insurance achieves its equity objectives alongside its access goals.

References

Ayer, Rakesh, Suresh Tiwari, Shambhu Prasad Jnawali, and Rudi Van Dael. 2024. *A Study on Nepal’s National Health Insurance Program* [in en]. Technical report TCS240257-2. Accepted: 2024-05-09 Last Modified: 2024-06-05T13:39+08:00. Asian Development Bank, May. Accessed November 28, 2025. <https://www.adb.org/publications/study-nepal-national-health-insurance-program>. (Cited on page 3).

- Banerjee, Abhijit, Amy Finkelstein, Rema Hanna, Benjamin A. Olken, Arianna Ornaghi, and Sudarno Sumarto.** 2021. “The Challenges of Universal Health Insurance in Developing Countries: Experimental Evidence from Indonesia’s National Health Insurance” [in en]. *American Economic Review* 111, no. 9 (September): 3035–3063. ISSN: 0002-8282, accessed April 23, 2024. <https://doi.org/10.1257/aer.20200523>. <https://www.aeaweb.org/articles?id=10.1257/aer.20200523&ArticleSearch%5Bwithin%5D%5Barticletitle%5D=1&ArticleSearch%5Bwithin%5D%5Barticleabstract%5D=1&ArticleSearch%5Bwithin%5D%5Bauthorlast%5D=1&ArticleSearch%5Bq%5D=health+insurance&JelClass%5Bvalue%5D=0&journal=&from=a&from=j>. (Cited on page 22).
- Borusyak, Kirill, Xavier Jaravel, and Jann Spiess.** 2024. “Revisiting Event-Study Designs: Robust and Efficient Estimation.” *The Review of Economic Studies* (February): rdae007. ISSN: 0034-6527, accessed May 5, 2024. <https://doi.org/10.1093/restud/rdae007>. <https://doi.org/10.1093/restud/rdae007>. (Cited on pages 11, 53).
- Callaway, Brantly, and Pedro H. C. Sant’Anna.** 2021. “Difference-in-Differences with multiple time periods.” *Journal of Econometrics*, Themed Issue: Treatment Effect 1, 225, no. 2 (December): 200–230. ISSN: 0304-4076, accessed May 5, 2024. <https://doi.org/10.1016/j.jeconom.2020.12.001>. <https://www.sciencedirect.com/science/article/pii/S0304407620303948>. (Cited on pages 2, 11, 14, 36, 48, 53).
- CBS.** 2011. *Nepal Living Standards Survey 2010-2011, NLSS Third [Data set]*. Central Bureau of Statistics (CBS), Nepal. Accessed December 1, 2021. <http://cbs.gov.np/nada/index.php/catalog>. (Cited on page 10).
- Chaisemartin, Clément de, and Xavier D’Haultfœuille.** 2020. “Two-Way Fixed Effects Estimators with Heterogeneous Treatment Effects” [in en]. *American Economic Review* 110, no. 9 (September): 2964–2996. ISSN: 0002-8282, accessed May 5, 2024. <https://doi.org/10.1257/aer.20181169>. <https://www.aeaweb.org/articles?id=10.1257/aer.20181169>. (Cited on pages 11, 53).

- Chetty, Raj, and Adam Looney.** 2006. “Consumption smoothing and the welfare consequences of social insurance in developing economies” [in en]. *Journal of Public Economics* 90, no. 12 (December): 2351–2356. ISSN: 00472727, accessed May 5, 2024. <https://doi.org/10.1016/j.jpubeco.2006.07.002>. <https://linkinghub.elsevier.com/retrieve/pii/S004727270600082X>. (Cited on page 2).
- Das, Jishnu, and Quy-Toan Do.** 2023. “The Prices in the Crises: What We Are Learning from 20 Years of Health Insurance in Low- and Middle-Income Countries” [in en]. *Journal of Economic Perspectives* 37, no. 2 (May): 123–152. ISSN: 0895-3309, accessed May 16, 2024. <https://doi.org/10.1257/jep.37.2.123>. <https://www.aeaweb.org/articles?id=10.1257/jep.37.2.123>. (Cited on page 2).
- Dupas, Pascaline, and Radhika Jain.** 2024. “Women Left Behind: Gender Disparities in Utilization of Government Health Insurance in India.” *American Economic Review* 114, no. 10 (October): 3345–85. <https://doi.org/10.1257/aer.20230521>. <https://www.aeaweb.org/articles?id=10.1257/aer.20230521>. (Cited on pages 27, 28, 39).
- Fitzpatrick, Anne, and Rebecca Thornton.** 2019. “The Effects of Health Insurance within Families: Experimental Evidence from Nicaragua.” *The World Bank Economic Review* 33, no. 3 (October): 736–749. ISSN: 0258-6770, accessed May 8, 2024. <https://doi.org/10.1093/wber/lhx012>. <https://doi.org/10.1093/wber/lhx012>. (Cited on page 2).
- GoN.** 2021. *Nepal Multidimensional Poverty Index 2021* [in en]. Discussion Papers. National Planning Commission, Government of Nepal. Accessed November 28, 2025. <https://www.undp.org/nepal/publications/nepal-multidimensional-poverty-index-2021>. (Cited on page 3).
- Goodman-Bacon, Andrew.** 2021. “Difference-in-differences with variation in treatment timing.” *Journal of Econometrics*, Themed Issue: Treatment Effect 1, 225, no. 2 (December): 254–277. ISSN: 0304-4076, accessed May 5, 2024. <https://doi.org/10.1016/j.jeconom.2021.03.014>. <https://www.sciencedirect.com/science/article/pii/S0304407621001445>. (Cited on page 11).
- Hanna, Rema, and Benjamin A Olken.** 2026. *The Handbook of Social Protection: Evidence and New Directions for Low-and Middle-Income Countries*. MIT Press. (Cited on page 32).

- HIB.** 2024. *Health Insurance Board Strategy Roadmap 2081/82* [in en]. Technical report. Kathmandu, Nepal: Department of Health Services: Health Insurance Board. Accessed March 28, 2025. https://hib.gov.np/public/uploads/shares/HIB_Strategy_Roadmap_2081_82.pdf. (Cited on page 4).
- Huang, Wei, and Hong Liu.** 2023. “Early childhood exposure to health insurance and adolescent outcomes: Evidence from rural China.” *Journal of Development Economics* 160 (January): 102925. ISSN: 0304-3878, accessed May 8, 2024. <https://doi.org/10.1016/j.jdeveco.2022.102925>. <https://www.sciencedirect.com/science/article/pii/S0304387822000815>. (Cited on page 2).
- Kurowski, Christoph, Anurag Kumar, Julio Cesar Mieses Ramirez, Martin Schmidt, and Denise Valerie Silfverberg.** 2023. *Health Financing in a Time of Global Shocks: Strong Advance, Early Retreat*. Publication Title: Health Financing in a Time of Global Shocks: Strong Advance, Early Retreat. World Bank, May. <https://doi.org/10.1596/39864>. (Cited on page 2).
- Miller, Grant, Diana Pinto, and Marcos Vera-Hernández.** 2013. “Risk Protection, Service Use, and Health Outcomes under Colombia’s Health Insurance Program for the Poor” [in en]. *American Economic Journal: Applied Economics* 5, no. 4 (October): 61–91. ISSN: 1945-7782, accessed April 23, 2024. <https://doi.org/10.1257/app.5.4.61>. <https://www.aeaweb.org/articles?id=10.1257/app.5.4.61>. (Cited on page 2).
- National Planning Commission.** 2015. *Nepal Earthquake 2015: Post Disaster Needs Assessment*. Kathmandu: National Planning Commission, Government of Nepal. https://www.npc.gov.np/images/category/PDNA_volume_BFinalReport.pdf. (Cited on page 15).
- National Statistics Office.** 2023. *National Population and Housing Census 2021 (National Report)* [in en]. Report. National Statistics Office, Government of Nepal. Accessed November 28, 2025. <https://censusnepal.cbs.gov.np/results/downloads/national?type=report>. (Cited on page 3).
- Rambachan, Ashesh, and Jonathan Roth.** 2023. “A More Credible Approach to Parallel Trends.” *Review of Economic Studies* 90 (5): 2555–2591. <https://doi.org/10.1093/restud/rdad018>. (Cited on page 49).

- Romano, Joseph P., and Michael Wolf.** 2005. “Stepwise Multiple Testing as Formalized Data Snooping.” *Econometrica* 73 (4): 1237–1282. <https://doi.org/10.1111/j.1468-0262.2005.00615.x>. (Cited on page 14).
- Roth, Jonathan, Pedro H. C. Sant’Anna, Alyssa Bilinski, and John Poe.** 2023. “What’s trending in difference-in-differences? A synthesis of the recent econometrics literature.” *Journal of Econometrics* 235, no. 2 (August): 2218–2244. ISSN: 0304-4076, accessed May 5, 2024. <https://doi.org/10.1016/j.jeconom.2023.03.008>. <https://www.sciencedirect.com/science/article/pii/S0304407623001318>. (Cited on pages 11, 13, 47).
- Sant’Anna, Pedro H.C., and Jun Zhao.** 2020. “Doubly robust difference-in-differences estimators.” *Journal of Econometrics* 219 (1): 101–122. ISSN: 0304-4076. <https://doi.org/https://doi.org/10.1016/j.jeconom.2020.06.003>. <https://www.sciencedirect.com/science/article/pii/S0304407620301901>. (Cited on page 14).
- WHO.** 2022. *Global Health Expenditure Database*. Accessed November 28, 2025. <https://apps.who.int/nha/database>. (Cited on page 3).
- WHO, and World Bank.** 2023. *Tracking Universal Health Coverage: 2023 Global Monitoring Report*. Eprint: <https://elibrary.worldbank.org/doi/pdf/10.1596/40348>. World Bank. <https://doi.org/10.1596/40348>. <https://elibrary.worldbank.org/doi/abs/10.1596/40348>. (Cited on page 2).
- World Bank.** 2020. *Household Risk and Vulnerability Survey, Full Panel 2016–2018 [Data set]*. World Bank, Development Data Group. Accessed December 1, 2021. <https://doi.org/https://doi.org/10.48529/mqys-cp36>. <https://doi.org/10.48529/mqys-cp36>. (Cited on page 10).
- . 2025a. *World Bank Development Indicators* [in en]. Accessed November 28, 2025. <https://data.worldbank.org>. (Cited on page 3).
- . 2025b. *World Bank Development Indicators* [in en]. Accessed November 28, 2025. <https://data.worldbank.org>. (Cited on page 3).

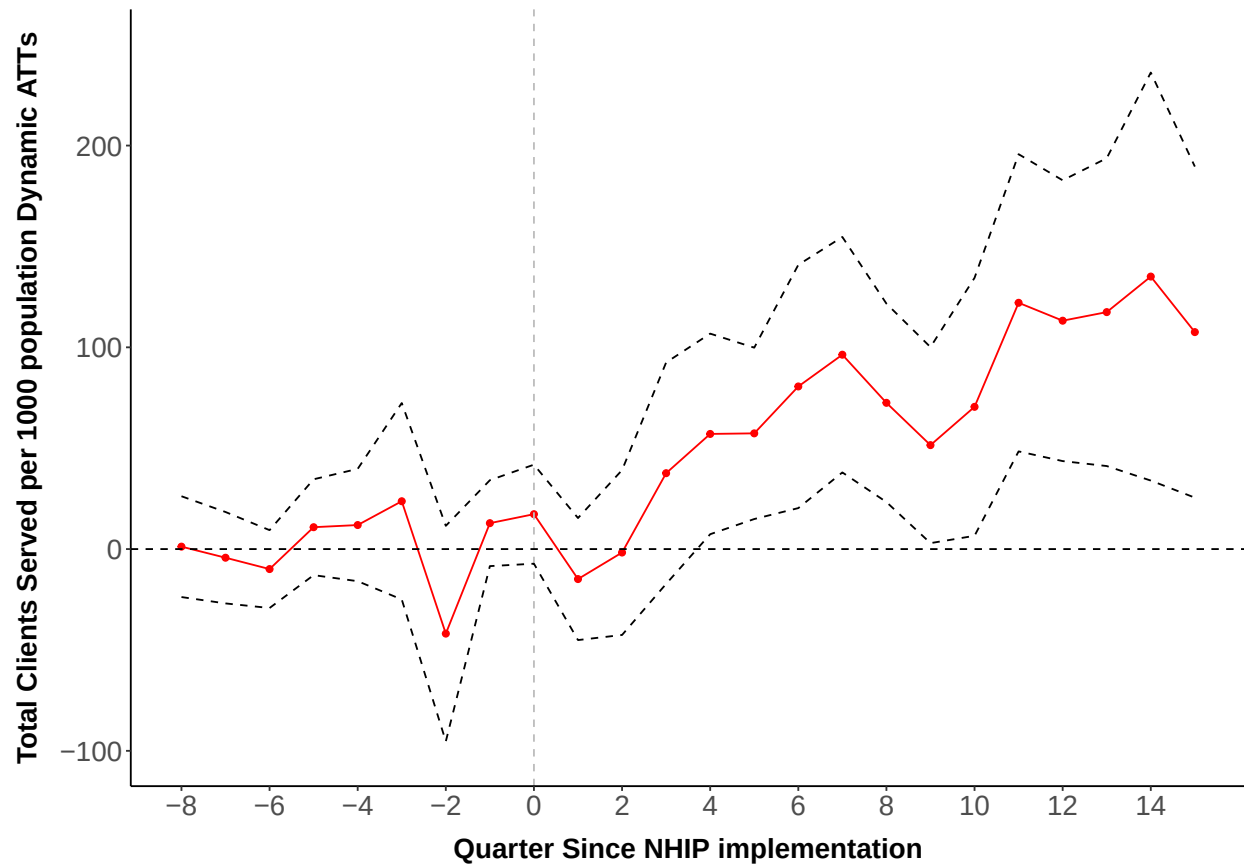
A — Appendix

Table A1 — Health Insurance Eligibility on Health Visits (per 1000 population)

	Total Clients Served	Total New Clients Served	Total OPD Visits	Total New OPD Visits
Model:	(1)	(2)	(3)	(4)
<i>Group Aggregation</i>				
Is NHIP eligible = 1	39.7522*** (7.3389)	17.5118 (9.2275)	38.0590*** (7.2238)	28.6267*** (5.6907)
Pre-Treatment Mean	267.71	222.04	239.28	199.67
<i>Fit statistics</i>				
Number of Cohorts	17	17	17	17
Number of Time Periods	30	30	30	30

Note: All the outcomes are per 1000 population. Clustered (District) standard-errors in parentheses bootstrapped with 1000 iterations. Estimation Method: Doubly Robust. Control Group: Not Yet Treated, Anticipation Periods: 1. Signif. Codes: *** 99% ** 95% * 90% uniform confidence band does not include 0.

Figure A1 – Total Clients Served per 1000 population Dynamic ATTs



Notes. This figure shows dynamic ATTs for total clients served per 1000 population by health service providers. The outcome is total clients served per 1000 population, x-axis shows weeks since NHIP implementation, quarter 0 being the first implementation quarter

A.1. Rollout Balance

Table A2 — Rollout Timing and Pre-Treatment Observables

	Coefficient	Std. Error
Avg. Total Visits (1000s)	0.0566	(0.0383)
Avg. OPD Visits (1000s)	−0.0161	(0.0343)
Avg. Emergency Visits (1000s)	−0.1556	(0.1921)
Population (1000s)	−0.0051	(0.0062)
No. of Hospitals	−0.0150	(0.0326)
Poverty Rate	−3.3813	(5.8969)
<i>Fit statistics</i>		
Observations	77	
R^2	0.0882	
Joint F-statistic	1.129	
F-test p-value	0.3545	

Note: This table reports OLS estimates from regressing the treatment timing variable (first quarter of NHIP implementation) on pre-treatment district characteristics. Pre-treatment means are computed over all quarters before treatment. The joint F-test evaluates whether observables jointly predict rollout timing. Failure to reject supports the assumption that rollout timing is as-good-as-random with respect to district characteristics.

A.2. Pre-Trends Test

Table A3 — Joint Test of Pre-Treatment Trends

	Wald χ^2	Degrees of Freedom	p -value
Total Visits	14.661	8	0.0661
OPD Visits	31.479	8	0.0001
Emergency Visits	13.719	8	0.0894
New OPD Visits	19.230	8	0.0137
New Emergency Visits	12.866	8	0.1165
New Total Visits	14.377	8	0.0725

Note: This table reports joint Wald tests of the null hypothesis that all pre-treatment dynamic ATT coefficients within 8 quarters (2 years) of treatment are simultaneously equal to zero. The test is restricted to the nearest pre-treatment periods to avoid over-rejection from an over-powered test across all 29 pre-treatment quarters. The test statistic follows a χ^2 distribution under the null. Failure to reject provides evidence consistent with parallel pre-treatment trends, though as Roth et al. (2023) emphasises, such tests have limited power against plausible violations.

A.3. Multiple Hypothesis Testing

Table A4 — Romano-Wolf Multiple Hypothesis Testing Correction

	ATT	Std. Error	Unadjusted p	Romano-Wolf p
Total Visits	0.1331***	(0.0234)	0.0000	0.0000
OPD Visits	0.1281***	(0.0307)	0.0000	0.0000
Emergency Visits	0.1790**	(0.0646)	0.0056	0.0186
New Total Visits	0.1052***	(0.0240)	0.0000	0.0000
New OPD Visits	0.1169***	(0.0358)	0.0011	0.0043
New Emergency Visits	0.1729**	(0.0638)	0.0067	0.0212
Laboratory Services	0.9580**	(0.3569)	0.0073	0.0212
Radiographic Services	0.1703**	(0.0698)	0.0147	0.0212
<i>Fit statistics</i>				
Number of Districts	77			
Hypotheses Tested	8			
Simulations	10,000			

Note: This table reports Romano-Wolf stepdown adjusted p -values that control the familywise error rate across all utilization outcomes tested simultaneously. The procedure uses the joint covariance structure of the test statistics, estimated via influence functions from the Callaway and Sant’Anna (2021) estimator, with 10,000 simulations. Significance stars reflect Romano-Wolf adjusted p -values. Signif. Codes: *** 1% ** 5% * 10%.

A.4. Sensitivity to Parallel Trends Violations

Figure A2 presents sensitivity analysis following Rambachan and Roth (2023). These plots show honest confidence intervals for the impact-period treatment effect ($e = 0$) under increasingly permissive violations of the parallel trends assumption using the relative magnitudes restriction (Δ^{RM}). The parameter $\bar{M}$ on the horizontal axis bounds the magnitude of post-treatment trend violations relative to the largest pre-treatment deviation. At $\bar{M} = 0$, parallel trends hold exactly; at $\bar{M} = 1$, post-treatment violations can be as large as the maximum observed pre-treatment deviation. The confidence intervals widen as $\bar{M}$ increases, indicating that the estimated treatment effects are sensitive to departures from exact parallel trends. This sensitivity underscores the importance of the complementary evidence supporting the identification strategy: pre-treatment coefficients are jointly insignificant, rollout timing is uncorrelated with pre-treatment observables (F -test $p = 0.35$), and results are robust to excluding earthquake-affected districts and truncating at the pre-COVID period.

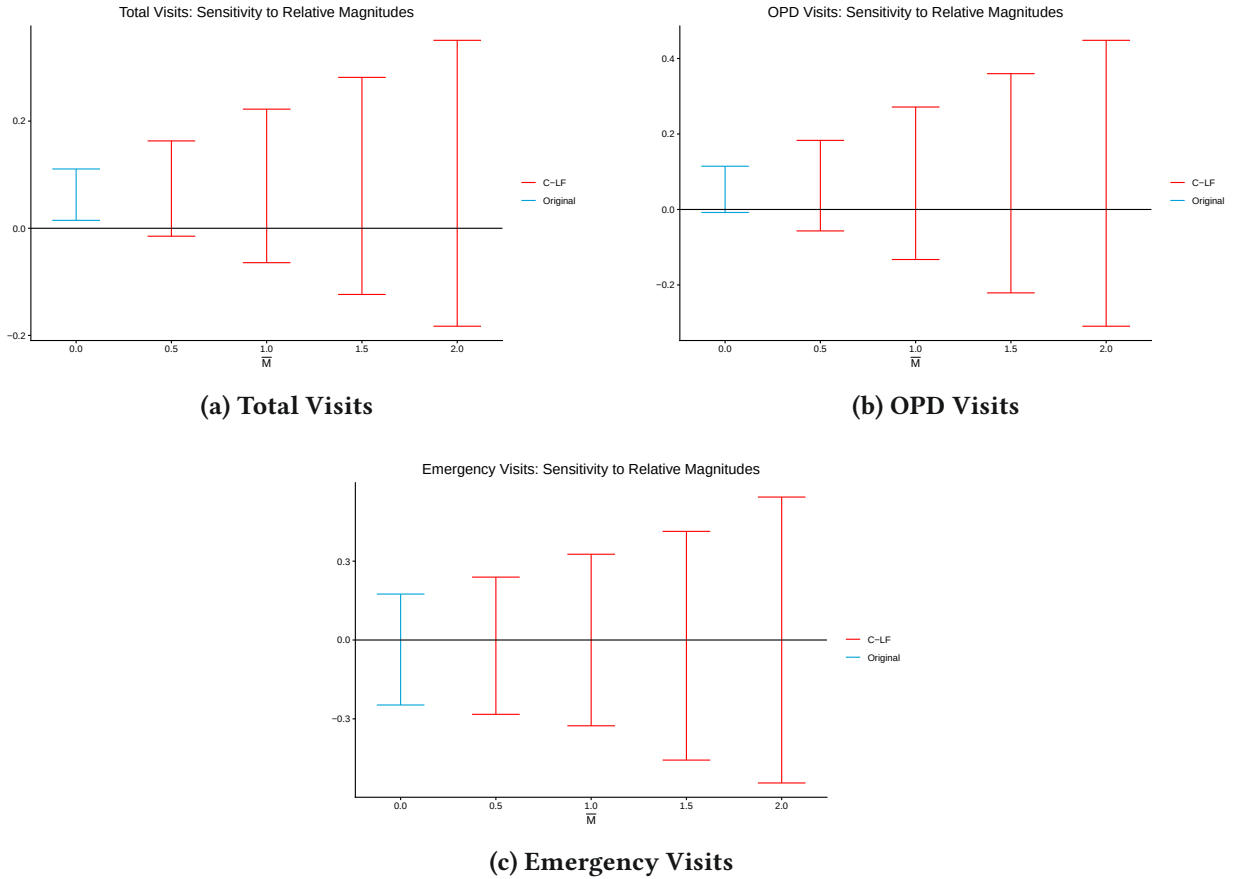


Figure A2 – Sensitivity of Treatment Effects to Parallel Trends Violations

A.5. Robustness Checks

Table A5 – Robustness: Excluding Earthquake-Affected Districts

	Estimate (Is NHIP eligible = 1)	Std. Error	Pre-treatment Mean
Total Visits	0.0675***	(0.0254)	79,262
OPD Visits	0.0327	(0.0417)	70,816
Emergency Visits	0.1245**	(0.0582)	5,582
<i>Fit statistics</i>			
Number of Districts	63		
Number of Cohorts	15		
Number of Time Periods	27		

Note: This table re-estimates the main specifications from [Table 2](#) Panel A after excluding the 14 districts severely affected by the 2015 Gorkha earthquake. All outcomes are log-transformed. Clustered (District) standard errors in parentheses. Signif. Codes: *** 1% ** 5% * 10%.

Table A6 – Robustness: Pre-Pandemic Sample (Through Q4 2019)

	Estimate (Is NHIP eligible = 1)	Std. Error	Pre-treatment Mean
Total Visits	0.0488**	(0.0234)	97,179
OPD Visits	0.0353	(0.0357)	75,396
Emergency Visits	0.0563	(0.0512)	8,274
<i>Fit statistics</i>			
Number of Districts	77		
Number of Cohorts	10		
Number of Time Periods	22		

Note: This table re-estimates the main specifications using only the pre-pandemic sample (through Q4 2019). All outcomes are log-transformed. Clustered (District) standard errors in parentheses. Signif. Codes: *** 1% ** 5% * 10%.

Table A7 — Robustness: Universal Base Period

	Estimate (Is NHIP eligible = 1)	Std. Error	Pre-treatment Mean
Total Visits	0.1331***	(0.0234)	101,980
OPD Visits	0.1281***	(0.0307)	78,357
Emergency Visits	0.1790***	(0.0646)	9,275
<i>Fit statistics</i>			
Number of Districts	77		
Number of Cohorts	17		
Number of Time Periods	30		

Note: This table re-estimates the main specifications using a universal base period instead of the varying (short gap) base period used in the main analysis. All outcomes are log-transformed. Clustered (District) standard errors in parentheses. Signif. Codes: *** 1% ** 5% * 10%.

Table A8 — Robustness: Public Facilities Only

	Estimate (Is NHIP eligible = 1)	Std. Error	Pre-treatment Mean
Total Visits (Public)	0.0712***	(0.0196)	72,972
OPD Visits (Public)	0.1122***	(0.0275)	58,849
Emergency (Public)	0.1776***	(0.0670)	4,899
<i>Fit statistics</i>			
Number of Districts	77		
Number of Cohorts	17		
Number of Time Periods	30		

Note: This table re-estimates the main specifications restricting the sample to public facilities only. All outcomes are log-transformed. Clustered (District) standard errors in parentheses. Signif. Codes: *** 1% ** 5% * 10%.

Table A9 – Robustness: Monthly Data with Varying Anticipation

	Estimate (Is NHIP eligible = 1)	Std. Error	Pre-treatment Mean
Panel A: Anticipation $\delta = 0$ months			
Total Visits	-0.0148	(0.0186)	35,998
OPD Visits	0.0144	(0.0254)	27,685
Emergency Visits	0.1437	(0.1106)	3,309
Panel B: Anticipation $\delta = 1$ months			
Total Visits	0.0398**	(0.0190)	35,998
OPD Visits	0.0338	(0.0248)	27,685
Emergency Visits	0.0899	(0.1196)	3,309
Panel C: Anticipation $\delta = 2$ months			
Total Visits	0.0956***	(0.0254)	35,998
OPD Visits	0.0860**	(0.0338)	27,685
Emergency Visits	0.1367**	(0.0570)	3,309
Panel D: Anticipation $\delta = 3$ months			
Total Visits	0.1639***	(0.0318)	35,998
OPD Visits	0.1452***	(0.0344)	27,685
Emergency Visits	0.1638***	(0.0580)	3,309
<i>Fit statistics</i>			
Number of Districts	77		
Number of Time Periods	91		

Note: This table re-estimates the main specifications using monthly rather than quarterly data, with varying anticipation periods ($\delta = 0, 1, 2, 3$ months). The main analysis aggregates to quarters with $\delta = 1$ quarter; this table verifies that results are robust to finer temporal resolution and different anticipation assumptions. All outcomes are log-transformed. Clustered (District) standard errors in parentheses. Signif. Codes: *** 1% ** 5% * 10%.

Table A10 – Robustness: Alternative Staggered DiD Estimators

	Total Visits	OPD Visits	Emergency Visits
Callaway & Sant’Anna (2021)	0.1331*** (0.0284)	0.1281*** (0.0353)	0.1790*** (0.0665)
de Chaisemartin & D’Haultfœuille (2020)	0.0819*** (0.0180)	0.0748*** (0.0262)	0.0782* (0.0447)
Borusyak et al. (2024)	0.1259*** (0.0229)	0.1101*** (0.0243)	0.2790* (0.1478)

Note: This table compares the main Callaway and Sant’Anna (2021) estimator with two alternative heterogeneity-robust staggered DiD estimators: the local-DID estimator of Chaisemartin and D’Haultfœuille (2020) and the imputation estimator of Borusyak, Jaravel, and Spiess (2024). All three estimators are designed to produce unbiased estimates under heterogeneous treatment effects in staggered adoption designs. All outcomes are log-transformed. Clustered (District) standard errors in parentheses. Signif. Codes: *** 1% ** 5% * 10%.

A.6. Distance Mechanism: Instrumental Variable Estimates

This section presents instrumental variable estimates complementing the reduced-form analysis in the main text. The FPOC reassignment indicator ($\text{Treated} \times \text{Post}$) instruments for distance to care; the first stage is reported in Table 7 and the second stage in Table A11. The instrument is strong for the full sample and low-poverty districts (first-stage $F > 1,000$), but weak for high-poverty districts ($F = 14.6$), consistent with the policy not significantly shifting distance in those areas.

Table A11 – IV Estimates: Effect of Distance on Insurance Claims

	Full Sample		High Poverty		Low Poverty	
	(1)	(2)	(3)	(4)	(5)	(6)
Distance (km)	-0.293***	-0.746***	0.336	-0.314*	-0.372***	-0.797***
	(0.098)	(0.168)	(0.208)	(0.181)	(0.127)	(0.194)
<i>Fixed effects</i>						
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes
Ward $\times$ Post	No	Yes	No	Yes	No	Yes
<i>Instrument: Treated $\times$ Post</i>						
First-stage F	13,860.2	7,195.2	4,344.2	2,057.2	10,519.3	5,544.9
<i>Summary statistics</i>						
Mean claims	4.04		3.63		4.15	
Mean distance (km)	7.6		7.4		7.7	
Observations	697,698		147,000		550,698	
N (individuals)	373,281		80,478		292,803	

Note: This table reports two-stage least squares estimates of the effect of distance (km) to FPOC hospital on the number of insurance claims per individual. Distance is instrumented by $\text{Treated} \times \text{Post}$, the interaction of FPOC reassignment with the post-2019 indicator. Odd columns include individual fixed effects; even columns add ward $\times$ post interactions. Poverty groups defined by median split of district poverty rate. The first-stage F statistic tests the relevance of the instrument; values below 10 indicate a weak instrument. Ward-clustered standard errors in parentheses. Distance outliers above 25km excluded. Signif. Codes: *** 1% ** 5% * 10%.

Note: 2SLS estimates. Dependent variable is number of insurance claims. Distance to FPOC (km) is instrumented by $\text{Treated} \times \text{Post}$. Individual fixed effects and ward-clustered standard errors in parentheses. First-stage F-statistics reported. Signif. Codes: *** 1% ** 5% * 10%.

A.6.1 Distance Robustness

Table A12 reports the first-stage and reduced-form estimates under alternative fixed effects specifications. Coarser unit-level fixed effects (VDC, ward) produce sign reversals, indicating that treated and untreated individuals differ systematically in baseline characteristics. Household fixed effects yield intermediate results. Individual fixed effects—the preferred specification—are necessary to absorb individual-level heterogeneity. The results are similar when replacing ward $\times$ post with VDC $\times$ post interactions.

Table A12 — Distance Analysis: Robustness to Geographic Time Interactions

	Full Sample	High Poverty	Low Poverty
Panel A: First Stage (Distance $\sim$ Treated$\times$Post)			
Individual FE + Ward $\times$ Post	-0.685*** (0.170)	-0.852** (0.411)	-0.659*** (0.186)
Individual FE + VDC $\times$ Post	-0.740*** (0.179)	-0.890** (0.404)	-0.716*** (0.198)
Individual FE + District $\times$ Post	-0.819*** (0.193)	-1.286** (0.540)	-0.742*** (0.207)
Panel B: Reduced Form (Claims $\sim$ Treated$\times$Post)			
Individual FE + Ward $\times$ Post	0.633*** (0.045)	0.256*** (0.052)	0.692*** (0.049)
Individual FE + VDC $\times$ Post	0.643*** (0.045)	0.149** (0.074)	0.721*** (0.049)
Individual FE + District $\times$ Post	0.569*** (0.054)	0.047 (0.152)	0.655*** (0.055)
Observations	611,275	140,093	471,182
N (individuals)	315,363	73,053	242,310

Note: This table reports robustness of the first-stage and reduced-form estimates to alternative geographic time interactions. All specifications include individual fixed effects; rows vary the geographic level of the time interaction from ward (1,810 units) to VDC (324 units) to district (53 units). The preferred specification (Ward $\times$ Post) corresponds to the main-text tables. Ward-clustered standard errors in parentheses. Poverty groups defined by median split of district poverty rate. Distance outliers above 25 km excluded. Signif. Codes:

*** 1% ** 5% * 10%.

Table A13 tests whether the results are driven by COVID-19 contamination. Since the post-period spells overlap substantially with the pandemic, I progressively restrict the sample to earlier enrollment cohorts with less COVID exposure. The reduced-form effect is largest for the cleanest

cohort (May 2019 only, $\sim 14\%$ COVID overlap) and declines gradually as later cohorts with greater COVID exposure are added. The ward $\times$ post fixed effects absorb ward-level pandemic shocks, and the stability of the estimates across cohort windows indicates that COVID contamination does not drive the main results.

Table A13 — Distance Analysis: Robustness to COVID Exposure

Enrollment Window	COVID%	Full Sample	High Poverty	Low Poverty	N (ind.)
Panel A: First Stage (Distance ~ Treated×Post)					
May 2019 only	~14	-0.681** (0.345)	-0.498* (0.295)	-0.706* (0.391)	41,022
May–Aug 2019	~27	-0.644** (0.264)	-0.419 (0.267)	-0.676** (0.300)	99,225
May–Nov 2019	~40	-0.642*** (0.214)	-0.597** (0.275)	-0.649*** (0.245)	145,516
May 2019–Feb 2020	~52	-0.556*** (0.180)	-0.624* (0.326)	-0.545*** (0.203)	191,279
All cohorts	~70	-0.685*** (0.170)	-0.852** (0.411)	-0.659*** (0.186)	305,937
Panel B: Reduced Form (Claims ~ Treated×Post)					
May 2019 only	~14	1.003*** (0.143)	0.324* (0.170)	1.096*** (0.156)	41,022
May–Aug 2019	~27	0.814*** (0.095)	0.251** (0.121)	0.897*** (0.105)	99,225
May–Nov 2019	~40	0.737*** (0.072)	0.153* (0.088)	0.833*** (0.078)	145,516
May 2019–Feb 2020	~52	0.654*** (0.059)	0.208*** (0.076)	0.725*** (0.065)	191,279
All cohorts	~70	0.633*** (0.045)	0.256*** (0.052)	0.692*** (0.049)	305,937

Note: This table reports robustness of the distance analysis to progressively wider enrollment cohort windows. Each row adds later-enrolling cohorts whose post-period spells have greater overlap with the COVID-19 pandemic (March 2020 onward). COVID% is the approximate share of post-period days falling after March 24, 2020. All specifications use individual FE + ward×post FE with ward-clustered standard errors. Distance outliers above 25 km excluded. Signif. Codes: *** 1% 5% * 10%.

Table A14 varies the maximum distance threshold used to define the analysis sample. The main specification uses 25 km. The reduced-form effect is stable across bandwidths from 10 to 40 km, ranging from 0.47 to 0.63 in the full sample, and is statistically significant at all thresholds. The first-stage distance effect is similarly robust.

Table A14 — Distance Analysis: Robustness to Distance Bandwidth

Bandwidth	Full Sample	High Poverty	Low Poverty	N (ind.)
Panel A: First Stage (Distance $\sim$ Treated$\times$Post)				
0–10 km	-0.125 (0.136)	-0.110 (0.169)	-0.128 (0.160)	236,981
0–15 km	-0.521*** (0.143)	-0.197 (0.162)	-0.577*** (0.164)	277,379
0–20 km	-0.572*** (0.166)	-0.291 (0.184)	-0.616*** (0.189)	302,752
0–25 km	-0.685*** (0.170)	-0.852** (0.411)	-0.659*** (0.186)	315,363
0–30 km	-0.725*** (0.171)	-0.939** (0.422)	-0.691*** (0.187)	323,264
0–40 km	-0.762*** (0.184)	-1.112** (0.502)	-0.704*** (0.198)	331,237
Panel B: Reduced Form (Claims $\sim$ Treated$\times$Post)				
0–10 km	0.469*** (0.064)	0.282*** (0.053)	0.508*** (0.075)	236,981
0–15 km	0.559*** (0.051)	0.284*** (0.050)	0.606*** (0.058)	277,379
0–20 km	0.624*** (0.048)	0.298*** (0.047)	0.675*** (0.054)	302,752
0–25 km	0.633*** (0.045)	0.256*** (0.052)	0.692*** (0.049)	315,363
0–30 km	0.623*** (0.042)	0.258*** (0.052)	0.682*** (0.047)	323,264
0–40 km	0.611*** (0.041)	0.272*** 60 (0.049)	0.668*** (0.046)	331,237

Note: This table reports robustness of the distance analysis to varying maximum distance thresholds. Each row restricts the sample to individuals whose FPOC distance falls below the stated bandwidth in both periods. The

A.6.2 FPOC Transition Heterogeneity and Monotonicity

A key identification concern is that the instrument does not satisfy monotonicity in the strict sense: the 2019 policy shifted distance in opposite directions for different FPOC transition types. [Table A15](#) shows the transition matrix for treated individuals. Rather than deterministically reassigning individuals from private to public hospitals, the policy reshuffled FPOC assignments in all directions, including 28.3% who moved from public to private facilities—the opposite of the policy’s stated intent. This reflects incomplete enforcement and the classification of NGO-run facilities as private.

Table A15 — FPOC Transition Matrix: Pre- to Post-2019 Hospital Assignment

Pre-2019 FPOC	Post-2019 FPOC		Total	Mean Δ dist. (km)
	Public	Private		
Public	17,843 (17.7%)	29,510 (29.3%)	47,353	—
Private	43,382 (43%)	10,149 (10.1%)	53,531	—
Total	61,225	39,659	100,884	

Mean distance change by transition (km)

Public → Public	-2.7 km
Public → Private	1.4 km
Private → Public	-1.9 km
Private → Private	-0.6 km

Note: This table shows the transition matrix of FPOC hospital ownership for treated individuals (those whose FPOC was reassigned by the 2019 policy restricting FPOC to public facilities). Hospital ownership classified from facility names; NGO facilities classified as private. Mean distance change is the mean of (post-2019 distance – pre-2019 distance) in km. Untreated individuals (not shown) have zero transitions by construction.

[Table A16](#) reports IV estimates separately by transition type. The public-to-public subsample—where there is no change in provider type and distance uniformly decreased—yields a significant IV coefficient of -0.443 (SE: 0.078, $F = 5,824$). The private-to-public subsample bundles distance

reduction with a provider-type change and yields a smaller coefficient (-0.141), consistent with opposing effects. Transitions where distance increased (public-to-private) show near-zero and insignificant effects.

Table A16 — IV Estimates by FPOC Transition Type

Transition	IV Coef.	Std. Error	Stage 1	F-stat	N (treated)	Obs.
Full sample	-0.746***	(0.168)	-0.854	7,195.2	125,316	697,484
public -> public	-0.437***	(0.054)	-2.595	46,218.5	17,843	531,386
private -> public	-0.393***	(0.080)	-2.458	86,691.1	43,382	582,466
public -> private	-0.004	(0.039)	1.655	34,675.0	29,510	554,724
private -> private	-0.510*	(0.260)	-0.555	8,307.9	10,149	515,996

Note: This table reports IV estimates of the effect of distance (km) on insurance claims, separately by FPOC transition type. Each row uses the indicated treated subsample plus all untreated individuals as the control group. Distance is instrumented by $\text{Treated} \times \text{Post}$. The public $\rightarrow$ public subsample isolates the distance channel with minimal provider-type confound. All specifications include individual FE and ward $\times$ post interactions. Hospital ownership classified from facility names (name-pattern matching + web search); NGO facilities classified as private. Ward-clustered standard errors in parentheses. Distance outliers above 25km excluded. Signif. Codes: *** 1% ** 5% * 10%.

The heterogeneity across transition types underscores why the full-sample IV (-0.692) cannot be interpreted as a well-defined LATE: the instrument moves different subpopulations in opposite directions, violating the monotonicity assumption required for the Imbens-Angrist framework. The public-to-public subsample provides the cleanest causal estimate of the distance effect, free from both provider-type confounds and monotonicity violations.

A.6.3 Hospital Quality and the Exclusion Restriction

A second concern is that FPOC reassignment simultaneously changed hospital quality, potentially confounding the distance channel. I construct a quality proxy using the number of unique ICD diagnosis codes observed at each hospital in the pre-2019 period, measuring service breadth and diagnostic capacity. [Table A17](#) shows that treated individuals were systematically reassigned to *lower*-quality hospitals: the mean quality change is -208 unique diagnoses, and over 60% of

treated individuals moved to hospitals with fewer diagnoses across all transition types. Yet the reduced form (Table 8) shows that claims *increased* by 0.59 on average. This rules out a “better hospital” channel and suggests that distance reduction is the dominant mechanism.

Table A17 – Hospital Quality Change: Pre-2019 Diagnosis Diversity

	Mean Quality	Mean Quality	Mean	% Assigned	
	Pre-2019	Post-2019	Change	Worse	N
<i>Panel A: By Treatment Status</i>					
Untreated	534.7	534.7	0	0%	247,343
Treated (all)	654.2	443.7	-210.5	65.8%	88,344
<i>Panel B: By Transition Type (Treated Only)</i>					
Public → Public	620.7	409.7	-211	68.4%	14,749
Private → Public	670.3	550.7	-119.6	57.9%	41,422
Public → Private	629.8	328.9	-300.8	74.1%	23,627
Private → Private	701.7	300.9	-400.7	76.6%	8,546

Note: Hospital quality is proxied by the number of unique ICD diagnosis codes observed at the hospital in the pre-2019 period, measuring service breadth. “Mean Change” is the mean of (post-2019 FPOC quality – pre-2019 FPOC quality). “% Assigned Worse” is the share of individuals whose post-2019 FPOC had fewer unique diagnoses than their pre-2019 FPOC. Untreated individuals have zero quality change by construction (their FPOC did not change). Hospital ownership classified from facility names; NGO facilities classified as private. The fact that treated individuals were assigned to lower-quality hospitals yet filed more claims (Table 8) indicates that the distance channel dominates the quality channel.

Table A18 provides additional evidence using alternative quality measures from the National Health Facility Registry (NHFR): facility level (primary/secondary/tertiary) and number of services offered. The three quality proxies give inconsistent directions across transition types, indicating that hospital quality does not systematically confound the distance-utilization relationship.

Table A18 — Robustness: Alternative Hospital Quality Proxies

	Mean Quality	Mean Quality	Mean	% Assigned	
	Pre-2019	Post-2019	Change	Worse	N
<i>Panel A: NHFR Facility Level (ordinal rank 1–13)</i>					
Untreated	7.25	7.25	0.00	0.0%	241,033
Treated (all)	7.23	7.43	0.20	37.6%	81,755
<i>By transition:</i>					
Public → Public	6.99	6.76	-0.23	36.9%	15,245
Private → Public	7.42	7.40	-0.02	45.7%	38,798
Public → Private	6.97	7.83	0.86	25.2%	21,829
Private → Private	7.54	7.89	0.35	32.2%	5,883
<i>Panel B: NHFR Services Count</i>					
Untreated	139.4	139.4	0.0	0.0%	163,325
Treated (all)	93.7	108.7	15.1	43.8%	19,198
<i>By transition:</i>					
Public → Public	150.6	166.2	15.6	47.5%	7,060
Private → Public	28.9	108.9	80.0	6.7%	6,062
Public → Private	124.8	55.9	-68.9	91.3%	4,455
Private → Private	2.2	2.8	0.6	35.8%	1,621

Note: This table replicates the quality change analysis from [Table A17](#) using two alternative quality proxies from the Nepal Health Facility Registry (NHFR). Panel A uses an ordinal facility level rank (1 = Basic Health Service Center to 13 = Super Specialized Hospital) based on the government’s official facility classification. Panel B uses the count of services listed in the NHFR registry as a measure of service breadth. Both proxies are independent of claims data. “% Assigned Worse” is the share of individuals whose post-2019 FPOC had a lower rank (Panel A) or fewer services (Panel B) than their pre-2019 FPOC. Untreated individuals have zero change by construction. Hospital ownership classified from facility names; NGO facilities classified as private. Coverage: Panel A has facility level for 463/501 hospitals; Panel B has services for 296/501 hospitals.

A.6.4 Attrition and Sample Selection

Since the balanced panel requires observation in both the pre- and post-2019 periods, individuals who stopped claiming after the policy change are excluded. If attrition were differential—higher among those more likely to be treated—the estimates would be biased. To test this, I classify all pre-2019 claimants with a recorded FPOC by their pre-period hospital type (public or private) and compare attrition rates. Table A19 shows that private-FPOC individuals, who were more likely to be reassigned by the policy, had a *lower* attrition rate (30.6%) than public-FPOC individuals (33.2%). A logistic regression confirms the difference is statistically significant after controlling for pre-period claiming intensity. This rules out differential attrition as a source of bias.

Table A19 — Attrition Test: Pre-Period Claimant Retention by FPOC Type

Pre-2019 FPOC Type	Pre-Period Claimants	Attrited (N)	Attrition Rate (%)	Mean Pre-Period Claims
Public	428,269	142,367	33.24	1.19
Private	177,485	54,381	30.64	1.24
Total	605,754	196,748	32.48	1.20

Logistic regression: Attrited ~ Private FPOC + Pre-Claims

Private FPOC	-0.0698*** (SE: 0.0062)
Pre-period claims	-1.0556*** (SE: 0.0083)

Note: This table tests for differential attrition from the balanced claims panel used in the distance analysis. The sample consists of pre-2019 claimants with a recorded FPOC hospital—the same population from which the balanced panel was drawn (individuals without a recorded FPOC were never candidates for the distance analysis). Pre-2019 claimants are classified by their modal (most frequent) pre-period FPOC hospital. “Attrited” indicates no post-2019 claims observed. Since treatment status (FPOC reassignment) cannot be observed for attriters, we proxy treatment exposure using pre-period FPOC type: individuals with private FPOCs were more likely to be reassigned by the 2019 policy. If the policy caused differentially higher attrition among those more likely to be treated, we would expect higher attrition among private-FPOC claimants. The logistic regression controls for pre-period claiming intensity. Hospital ownership classified from facility names; NGO classified as private.